Development of a Hydrogel Extruder

Mechatronic Project 478
Final Report

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Executive summary

Title of Project
Development of a Hydrogel Extruder
Objectives
To design and construct a syringe-based hydrogel extruder and 3-axis gantry capable of fabricating three-dimensional structures of moderate complexity.
What is current practice and what are its limitations?
Hydrogel bioprinters are currently used for tissue engineering and biomedical research, but existing printers are limited by slow curing rates, poor print resolution, and restricted control over extrusion parameters.
What is new in this project?
A modular hydrogel printer capable of controlled extrusion of multiple types of hydrogel bioinks and integrated curing, calibrated on a test gel to prove the concept.
If the project is successful, how will it make a difference?
The printer provides researchers with a versatile research tool to explore and fine-tune printing parameters for different hydrogels. It will contribute new knowledge to hydrogel 3D printing methods and support biomedical applications such as tissue scaffolding and cell culture environment fabrication.

What are the risks to the project being a success? Why is it expected to be successful?
Risks include budget and time constraints, and challenges in sourcing an affordable, accessible test hydrogel. Success is expected through a clear design methodology, calibration with surrogate materials, and careful documentation.
What contributions have/will other students made/make?
This is the second iteration of the project. Current work has established a functional prototype and experimental calibration framework. Future students can extend the design with advanced curing methods and dual extrusion capabilities.
Which aspects of the project will carry on after completion and why?
The use of the printer will continue in Stellenbosch University's Physiology Department, where it will be used as a tool for characterising different types of bioinks.
What arrangements have been/will be made to expedite continuation?
Comprehensive documentation of mechanical, electronic, and firmware design has been produced, along with an experimental calibration methodology, enabling seamless future development and integration into research environments.

ECSA self-assessment

GA 1. Problem solving
The project required problem solving to overcome multiple challenges in producing accurate and repeatable hydrogel prints. These included achieving controlled and stable extrusion of gels with complex non-Newtonian rheological properties, ensuring the system was adaptable to a range of bioinks, and providing a method to determine optimal printing parameters for different inks. Problem-solving ability is demonstrated in the mechanical, electronic, firmware, and experimental design aspects of the project.
GA 2. Application of scientific and engineering knowledge
The project required the integration of knowledge across mechanics, electronics, biology, and experimental design to develop a functional hydrogel printer. Mechanical analysis was used to calculate buckling loads in the syringe and to size actuators by determining the forces and speeds required for each stepper motor. Fluid mechanics principles, including Poiseuille's law, were applied to estimate extrusion pressures and inform the syringe design. Circuit analysis and embedded systems knowledge were applied in designing and programming the custom PCB for control and sensing. Biological understanding of tissue engineering provided the performance requirements that shaped the system's capabilities, while experimental design ensured validation of the printer's capabilities.
GA 3. Engineering Design
Engineering design was a major aspect of the project and was carried out across three primary phases: mechanical, embedded system, and experimental design. The mechanical design focused on developing the syringe-based extruder and gantry; the embedded system design centred on creating a custom PCB to control motion and sensing; and the experimental design established a methodology for determining the optimal printing parameters of a chosen bioink. Together these phases demonstrate a structured, multidisciplinary approach to achieving a functional bioprinter.

GA 5. Engineering methods, skills and tools, including Information Technology
The project required a wide range of engineering methods, skills and tools to design, build, and experimentally evaluate the 3D printer. Inventor Professional was used to create a 3D model of the mechanical bioprinter design. Analytical methods were applied to calculate mechanical loads, extrusion pressures, and motor requirements. Embedded hardware was designed using KiCad and the firmware was created through STM32CubeIDE. Experimental methods, supported by data processing in MATLAB, were applied to test the system and optimise printing parameters.
GA 6. Professional and technical communication
The project outcomes were communicated through an oral presentation and a detailed written report, supported by engineering drawings, schematics, and results. These demonstrated the ability to communicate information regarding technical work to engineering audiences.
GA 8. Individual, Team and Multidisciplinary Working
The project demonstrated independent execution while drawing on knowledge from multiple disciplines, including mechanics of materials, fluid mechanics, electronics, embedded software development, and biology. Collaboration with Stellenbosch University's workshop artisans and the Physiology Department ensured the system was manufacturable and aligned with clear design objectives.
GA 9. Independent Learning Ability
Independent learning was demonstrated in several aspects of the project such as the literature review, which provided foundational knowledge in tissue engineering and hydrogel material science, as well as exposure to new approaches in mechanical design. New skills were gained in PCB design using KiCad, which enabled the development of the bioprinter's circuit board for sensing and control.

Acknowledgements

I thank my Lord for sustaining me through this year and throughout my engineering degree, and for giving me the strength to complete my final-year skripsie project.

I am grateful to my supervisor, Mr. Swart, for his guidance throughout this project and for helping me broaden my range of engineering skills.

Lastly, I wish to thank my parents for their continuous support across my degree, and Stacey Sandy for her unwavering support and encouragement during my most academically challenging year.

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Constants

$\pi = 3.1416...$

$g = 9.81\ m \cdot s^2$

Variables

μ	Viscosity	$[\mathrm{Pa} \cdot \mathrm{s}]$
Q	Volumetric flow rate	$[\mathrm{mm}^3 \cdot \mathrm{s}^{-1}]$
v	Extrusion rate	$[\mathrm{mm} \cdot \mathrm{s}^{-1}]$
P	Pressure	$[\mathrm{Pa}]$
N	Force	$[\mathrm{N}]$

Subscripts

atm	Atmospheric
SF	Safety Factor

Abreviations

SU	Stellenbosch Univerisity
GelMA	Gelatin Methacrylate
PEGDA	Polyethylene Glycol Diacrylate
PVA	Polyvinyl Alcohol
EM	Extracellular Matrix
UV	Ultraviolet
CAD	Computer-Aided Design
PCB	Printed Circuit Board
PSU	Power Supply Unit
SWD	Serial Wire Debug
ISR	Interrupt Service Routine

Chapter 1

Introduction

1.1 Background

Tissue engineering is a relatively new field within the multidisciplinary realm of biomedical engineering. It combines the fields of biology, medicine, material science, and engineering into a single discipline in which three-dimensional structures are fabricated with the intention of restoring, maintaining, or even improving the function of tissue. The art of engineering tissue has many applications within medicine, cosmetics, and cellular research, and there are many methods of fabricating these tissue scaffolds, with one common approach being 3D printing.

The ability to create structures that replicate tissue functions, whether seeded with cells or not, has countless applications within various fields such as medicine, cosmetics, and biological research. It creates the potential to manufacture substitute tissues (or even organs in more advanced applications) in regenerative medicine for transplantation to mimic the function of damaged or missing tissues in patients. It is often used in drug development and testing. Tissue engineering is also used in biological research, where 3D environments are created to simulate and study the growth and behaviour of cellular systems such as cancer cells and bacteria.

A popular class of materials used to fabricate substitute tissue structures are hydrogels, which can be processed in various ways, with 3D printing being a particularly popular method. Hydrogels are a broad class of materials that are known for their tunable properties, porous nature, and ability to retain large amounts of water, which allows many hydrogels to support cell growth. However, due to the complex rheological properties of different hydrogels and the varying methods by which these materials are hardened, it remains a challenge to 3D print complex structures in a way that is accurate, repeatable, and cost-effective. This project aims to design, develop, and test an extrusion system capable of creating 3D hydrogel scaffolds to overcome these challenges.

1.2 Objectives

The primary objectives of this project are:

1. To design and assemble a hydrogel extruder capable of controlled and stable extrusion.

2. To design and construct a three-axis gantry to position the extruder for accurate 3D printing.

3. To design and implement an embedded system, including necessary firmware, capable of real-time control of system actuators, sensing, and closed-loop logic.

4. To experimentally tune and validate the mechatronic system through a designed calibration procedure for determining optimal printing parameters for different bioinks.

The scope of the project is limited to developing and demonstrating a working prototype. While the system is not intended to achieve the full performance and quality of commercial bioprinters, it provides a validated platform for future upgrades and serves as a tool for characterising bioinks.

1.3 Motivation

The motivation for this project lies in the growing importance of tissue engineering and the need for accessible tools to support research in this field. Most commercial bioprinters remain prohibitively expensive and proprietary, limiting their availability to smaller research groups. Developing a low-cost prototype capable of producing controlled, stable, and repeatable prints allows research in tissue engineering to be conducted by a wider range of researchers. This project therefore contributes to making tissue engineering more affordable and while providing a means of characterising different gels, offering insight into their behaviour in both tissue engineering and other biomedical applications.

1.4 Use of AI

The use of AI tools in this project was limited to identifying and correcting minor technical writing errors, including grammar, spelling, and punctuation. All technical aspects of the project, including the literature review, design, and experimental analysis, were independently performed.

Chapter 2

Literature review

2.1 Tissue engineering

2.1.1 Definition and Background of Tissue Engineering

Tissue engineering can be defined as the "the application of principles and methods of engineering and life sciences toward fundamental understanding of structure–function relationships in normal and pathological mammalian tissues and the development of biological substitutes to restore, maintain, or improve tissue function" (Skalak and Fox, 1988).

The concept of replacing dysfunctional tissues and organs has been around for millennia, with the first nose transplants occurring as early as 1000 B.C. in ancient India (Saltzman, 2004). In recent centuries, major advances in organ transplantation have taken place – even transplants of organs supplied by donors. These strides include tooth transplants, successful skin grafts (which became commonplace in the 1800s), and even the first successful heart transplant which took place in South Africa, 1967 (Saltzman, 2004).

Despite the rapid advancements in organ transplantation in recent centuries, all transplants involving organs from donors have a common issue. The problem is that the tissue/organ in need of replacement requires the availability of viable tissues or organs from a donor and in modern times the demand for tissue/organ replacement far exceeds the available supply from donors (Saltzman, 2004). This problem can be alleviated using tissue engineering.

Through tissue engineering, the demand for useable and available donor tissues is mitigated by creating viable tissues (and even organs) through artificial means. This replacement tissue is formed by creating a three-dimensional scaffold to take on the shape of the space of the human (or organism) that needs to be filled with new tissue. These scaffolds can be acellular or seeded with cells prior to formation. The acellular scaffolds rely on the body's natural ability to regenerate the cells for normal function. Whether seeded or regenerated from the body, the scaffolds (with its contained cells) are intended to replace the functionality of the old or damaged tissue (Olson *et al.*, 2011).

2.1.2 Scaffold Requirements

The main purpose of artificial scaffolding in tissue engineering is to mimic the environment and behaviour of the tissue being replaced. To create artificial scaffolding that recreates these same conditions, the scaffolding needs to be biocompatible, biodegradable, have sufficient architecture and have the same mechanical properties as the native extracellular matrix (O'Brien, 2011).

The scaffolds must be biocompatible with the cells. This means that the cells should be able to migrate through the scaffold material and be able to function normally within this artificial EM (extracellular matrix). This criterion also means that the engineered tissue should result in minimal immune response from the body once it is integrated (or implanted) with the host tissue (O'Brien, 2011).

The artificial scaffolds must be biodegradable since the scaffolds are not meant to be permanent implants. The scaffolds are intended to be gradually broken down and slowly replaced with a new extracellular matrix that is formed by the seeded or host's natural cells. To ensure that the tissue continues to function normally and remain undamaged, the by-products of the scaffold's degradation must be non-toxic (O'Brien, 2011).

The architecture of the scaffolds should be sufficient to allow for the cells to exist inside the matrix. This means that it should have an interconnected porous structure, and it should be highly porous. The reason for this, is so that the structure can allow for the diffusion of nutrients to the cells and the diffusion of waste products from the cells through the EM (O'Brien, 2011).

The scaffold's mechanical properties should match that of the native extracellular matrix or that of its intended purpose. This involves selecting materials (for scaffolding) that have similar stiffness (Young's modulus), hardness, etc. The importance of matching mechanical properties to the scaffolding can differ depending on what the engineered tissues intended purpose is. For example, if its function includes a structural role (as would be the case with bones and cartilage), then it is important to ensure it has enough stiffness and strength to not break upon usage. It should also be noted that the artificially formed matrix should have sufficient mechanical properties for it to remain undamaged when being implanted. In addition to this, it is important to ensure that the scaffold's mechanical properties do not get chosen at the cost of the scaffold porosity as this can result in the engineered tissue having difficulty in allowing vascularization and cell infiltration resulting in conditions where the cells cannot exist over time (O'Brien, 2011).

2.1.3 Scaffold Fabrication Methods

There are many methods of fabricating scaffolds. These techniques for creating scaffolds differ according to the material used to create the scaffold and depending on whether the seeded cells can survive the scaffolding formation process. The formation process also needs to ensure that the produced scaffolds meet the requirements for being used as engineered tissue (it must be biocompatible, porous, etc.). These varying methods can be split up into conventional and advanced methods (Dutta *et al.*, 2017).

Conventional techniques include particulate-leaching, extrusion, molding, thermally induced gelation, gas foaming, etc. (Dutta *et al.*, 2017). These processes can be used in combination to manufacture scaffoldings that meet the previously identified scaffold criteria and can work together to form more advanced manufacturing techniques.

Advanced methods include 3D printing, electrospinning, emulsion templating, and designed self-assembling peptides:

- 3D printing allows for the computer aided design of scaffolding and then the formation of these designs through additive manufacturing methods such as Fused Deposition Modelling, Stereolithography, and Laser Sintering; this style of fabrication is particularly useful for rapid prototyping, developing scaffolds with complex geometries, and having control over the macroscopic properties of the artificial matrix structure (Dutta *et al.*, 2017).

- Electrospinning involves using an extruder with an electric field to produce ultrafine nanofibers to form the extracellular matrix; this method is very useful for creating scaffolds with fibrous nanoporous materials with high precision (Dutta *et al.*, 2017).

- Emulsion templating uses phase separation to form tertiary pores to create scaffolds with a hierarchical pore structure; emulsion templating is useful for creating scaffolds with a porosity gradient (Dutta *et al.*, 2017).

2.1.4 Scaffold Materials

The material used to form the extracellular matrix depends on the application of the tissue that is being engineered. These possible materials include linear aliphatic polyesters and other synthetic polymers, natural macromolecules, inorganic materials, and hydrogels (Ma, 2004).

Linear aliphatic polyesters, such as PGA and PLA, are a group of biodegradable plastics that break down due to the hydrolysis of ester bonds. There are also other synthetic materials such as PPF and tyrosine-derived polymers that are used in bone engineering (Ma, 2004).

Natural macromolecules, like proteins polysaccharides, are often used in tissue engineering. Examples of fibrous proteins that are used to create tissue is collagen and silkworm silk. Collagens have useful properties and is therefore often used as a major component of EM scaffolding although it has potential issues with pathogen transmission and less controllable biodegradability; silk (from silkworms) is a useful tissue replacement option in certain instances due to its desirable tensile properties but has issues with cytotoxicity and slow degradation. Polysaccharides such as alginate, chitosan, and hyaluronate (these examples are or can be used to create hydrogels) are also useful in creating porous solid-state scaffolds (Ma, 2004).

Inorganic materials that fall into the categories of porous bioactive glasses and calcium phosphates are used in the fields of bone and mineralized tissue engineering as they support the adhesion, growth, and differentiation of cells (Ma, 2004).

Hydrogel polymers (like PEGs, alginates, etc.) are also a desirable option for producing extracellular scaffolds because they can be easily made to form structures of complex shapes/geometries, they can be seeded with cells, have solidification rates that are controllable (to some degree) and can often be implemented through procedures that are minimally invasive (Ma, 2004).

2.2 Hydrogel

2.2.1 Introduction to Hydrogels

Hydrogels are three-dimensional polymeric networks that are hydrophilic and therefore capable of containing large quantities of water while keeping their structural integrity (Peppas *et al.*, 2000). Hydrogels have many desirable properties that make it useful in many applications (even outside of tissue engineering).

The network structure of hydrogels is comprised of homopolymer or copolymer chains. These chains attract water, but the overall polymer network is still insoluble due to chemical and/or physical (entanglements, crystallites, etc.) crosslinks. Crosslinks are the tie points where the adjacent or neighboring chains are joined together. These crosslinks allow for the network structures to be solid and have some degree of structural integrity (Peppas *et al.*, 2000).

Hydrogels can be used in many applications. Due to their resemblance to the properties of natural extracellular matrices of tissue (as they are porous, enzymatically degradable, soft in consistency, biocompatible, and have high water content), they are an attractive option for applications within the medical and pharmaceutical sectors. Their biocompatibility (contributed to by its capacity to hold large amounts of water) allows it to make contact lenses, biosensor membranes, artificial heart lining, artificial skin materials, and drug delivery systems (Peppas *et al.*, 2000).

2.2.2 Classification of Hydrogels

Hydrogels can be classed in many ways and there are many kinds of hydrogels with different applications within the fields of medicine or tissue engineering.

Hydrogel can be classified according to its polymer source, physical properties, ionic charge, biodegradability, or its method of crosslinking. With regards to classifying these gels according to their source, they can either be natural, synthetic, or hybrid polymer hydrogels (Li *et al.*, 2020).

Natural hydrogels are a form of natural polymer (or biopolymer) as they are derived from organisms. These can be further classed into polysaccharide hydrogels (such as alginate which comes from brown algae), glycosaminoglycan hydrogels, and polypeptide/protein hydrogels (the most common example of these include collagen and gelatin hydrogels) (Li *et al.*, 2020).

Synthetic hydrogels are artificially created and come in the form of polyacrylamide (PAAm), poly(ethylene glycol) (PEG), and poly(vinyl alcohol) (PVA) hydrogel. PEG derived gels are the most used synthetic hydrogels in medicine due to their hydrophilic nature and excellent biocompatibility (Li *et al.*, 2020).

Hybrid hydrogels are made from a combination of natural and synthetic materials. The reason for this is that synthetic and natural gels on their own tend to have limited mechanical strength but through the combination of natural and synthetic materials this problem is rectified to allow improved and tunable mechanical properties as well as allow for the addition of other desirable characteristics to the gel product (Li *et al.*, 2020).

2.2.3 Properties of Hydrogels

Since there are countless types and forms of hydrogel, in this section, the properties of specific examples of hydrogels, relevant to the context of extrusion-based 3D printing with the intention of engineering tissue, are considered. These examples are GelMA (gelatin methacrylate), PEGDA (poly(ethylene glycol) diacrylate), Alginate, and Matrigel.

- Biocompatibility – all four of these hydrogels are highly biocompatible but only some are naturally able to adhere to cells. GelMA and Matrigel support cell adhesion; PEG-based hydrogels and Alginate naturally have poor cell adhesion, and this can be rectified by incorporating cell-adhesive peptides (Li *et al.*, 2020).

- Biodegradability – GelMA, Alginate, and Matrigel are enzymatically degradable. If the necessary enzymes are not available to degrade Alginate, it can also be ionically degraded while immersed in an aqueous solution with Na^+ ions. PEG-based gels are not degradable and need to be chemically modified to become biodegradable (Li *et al.*, 2020).

- Water retention – PEGDA, GelMA, and Alginate are all very soluble (in water and other solvents) and capable of holding large amounts of water (Li *et al.*, 2020). Matrigel needs to be chilled for it to be more easily soluble (Merceron and Murphy, 2015).

- Crosslinking method – PEGDA and GelMA (with a photoinitiator) are made to cure via photo crosslinking when irradiated by UV rays. Alginate is ionically crosslinked by divalent cations like Ca^{2+} (Li *et al.*, 2020). Matrigel, however, is thermally crosslinked (which is reversable). It is a liquid at low temperatures (around 4 °C), and it forms a solid matrix at higher temperatures of around 37 °C (Merceron and Murphy, 2015).

- Rheological properties – the rheological properties of hydrogels often depend on its temperature, concentration is a solute, and molecular weight. Before gelation, PEGDA is found to be a Newtonian fluid (to a certain shear flow rate) and its viscosity increases as its molecular weight increases (Brikov *et al.*, 2016). Both GelMA and Alginate, on the other hand, are non-Newtonian fluids that exhibit shear-thinning behaviour, meaning that their viscosity reduces as its shear rate increases (Gregory *et al.*, 2022). The viscosity of GelMA also varies depending on temperature and concentration. Its viscosity decreases as its temperature increases and its concentration decreases. The viscosity of GelMA, therefore, is very low when flowing at temperatures above 30 °C (significantly less than 1 $\mathrm{Pa \cdot s}$), but at cooler temperatures (less than 30 °C) its viscosity drastically increases (Adhikari *et al.*, 2021). For example, a solution with the GelMA concentration of 10% at 26 °C has a viscosity of roughly 168 $\mathrm{Pa \cdot s}$, which is a very large increase in viscosity (CELLINK, 2020).

2.3 3D Printing Hydrogel

As extrusion-based 3D printing has continued to grow as a method of engineering tissues with hydrogel, many companies have developed state-of-the-art hydrogel 3D printers for biomedical applications. The most advanced and notable bioprinters of today include the CELLINK BIO X printers, the RegenHU 3DDiscovery™ Evolution, and the Allevi 3 printer. To get an idea of the state of the art of hydrogel printers, this subsection will discuss the capabilities of the BIO X printer.

The BIO X 3D printer is developed by a company called CELLINK. It is capable of extruding and printing with many kinds of bioinks including GelMA, PEGDA, Alginate, etc. for application in biomedical research and has a print resolution of 1 µm (CELLINK, 2025).

The BIO X has interchangeable and attachable printheads for different forms of gel extrusions including a pneumatic, syringe pump, thermoplastic and inkjet printheads. The printing bed and printheads are temperature controlled (to account for the wide-ranging rheological properties of different hydrogels). The printers also include UV LED attachments for photo-induced crosslinking and implement a UV ray-based system (with smaller wavelengths than for curing) for sterilizing the printing environment in between prints to ensure biosafety standards are met (CELLINK, 2025).

Chapter 3

Mechanical Design

This chapter covers the mechanical design of the hydrogel bioprinter using the standard engineering design process that involves the identification of both stakeholder and engineering requirements, concept generation and selection, and the final design. The 3D printer system must fulfil two primary functions:

- Extrude hydrogel (specifically GelMA).
- Position the extruder in 3D space.

3.1 Stakeholder Requirements

The development of the hydrogel bioprinter is performed in accordance to the purpose of its use and the needs of its stakeholders. The 3D printer is being developed to serve as a research tool for SU's Physiology Department, in which it will be used as an apparatus for characterising bioinks and creating hydrogel scaffolds (acellular or seeded with cells) to model and simulate cellular growth in biomedical research. The stakeholders whose needs guide the design of the system include the physiology researchers (end-users), the engineering student developing the printer, regulatory authorities that require compliance with health and safety standards as well the strict sanitary standards that need to be met in biomedical research, and Stellenbosch University by which the project is funded. While not strictly a mandatory requirement, the system should also meet the needs of any future designer that is tasked with upgrading the system or adding additional functionalities to the printer. The requirements that the system must/should meet in order to fulfil the needs of these stakeholders are tabulated below in Table 3.1.

Table 3.1: Stakeholder Requirements

No.	Stakeholder	Description	Priority
SR1	Researchers (End-users)	The system must be able to extrude hydrogel to create 3D hydrogel scaffolds	must-have
SR2	Designer (Engineer)	The printer should be able to precisely, accurately, and smoothly adjust the position of the extruder in 3D space[1]	must-have
SR3	Designer (Engineer)	The extruder must be able to accurately and precisely control the flow rate of hydrogel being extruded[1]	must-have
SR4	Researchers (End-users)	The system should support the real-time gelation of the bioink during print tasks[2]	must-have
SR5	Researchers (End-users)	The bioprinter should support multiple types of hydrogel[3]	must-have
SR6	Researchers (End-users)	The printer should have a repeatable performance for unique set of parameters[1]	must-have
SR7	Researchers (End-users)	The printer must allow for tunable parameters[3]	must-have
SR8	Researchers (End-users)	The system should be user-friendly and simple to operate[4]	must-have
SR9	Biosafety Authorities	The extruder must be sterilisable and come with a documented cleaning procedure[5]	must-have
SR10	Health and Safety Authorities	The system should be safe to handle with minimized risks to the health and safety of its users	must-have
SR11	SU's finance department	The printing system as a whole should be affordable and meet budget requirements[6]	must-have
SR12	Researchers (End-users)	The system must include a documented experimental procedure to calibrate the bioprinter to perform optimised prints of desired bioinks[3]	must-have
SR13	Researchers (End-users)	The system should be able to transmit real time experimental data[3]	optional

Table 3.1: Stakeholder Requirements

No.	Stakeholder	Description	Priority
SR14	Designer (Engineer)	The printer should include feedback control for precise operation	optional
SR15	Future Designers	The printer should be modular and allow for future upgrades[7]	optional

[1] The calibrated printer should perform prints that accurately match the desired print tasks and do not vary across iterations with a specific set of printer settings.
[2] This means that the printer should support the solidification of the gel during print operations to ensure the structural integrity of printed scaffolds.
[3] These requirements are needed for the system to fulfil its role in serving as a tool for characterising bioinks and calibrating the printer to perform optimised prints.
[4] The end-users are not engineers and the printer should therefore be simple to operate by researchers without technical knowledge of mechatronic systems.
[5] The printer must be cleaned and sterilised throughout its lifetime to ensure bioink viability and compliance with health standards.
[6] This excludes resources that will be sourced from within the University of Stellenbosch.
[7] Future upgrades may include adding a user interface or developing an additional extruder for dual extrusion.

3.2 Engineering Requirements

The needs of the stakeholders must be translated into engineering requirements that define how the system will meet these needs.

Tables 3.2 and 3.3 list the FRs and NFRs respectively.

Table 3.2: Functional Requirements

No.	Description	Related SR
FR1	The printer must be able to move the extruder along three axes (relative to a printing platform) with precise positional control.	SR2, SR6
FR2	The system must extrude hydrogel at a desired and controlled flow rate.	SR1, SR3
FR3	The printer must have the ability to include UV curing with adjustable intensity that operates during extrusion.	SR4

No.	Description	Related SR
FR4	The printer must include the ability to adjust bioink fluid properties by bed and nozzle temperature control.	SR7, SR9
FR5	The printer firmware must allow for the adjustment of printer parameters such as nozzle and bed temperature, extrusion speed, and travel speed.	SR7
FR6	The system should be able to transmit sensory feedback for system control and to capture real-time data during the printer's calibration.	SR13, SR14
FR7	The printer must include feedback control for precise operation.	SR14

Table 3.3: Non-Functional Requirements

No.	Description	Related SR
NFR1	The extruder/bioink fluid path must be removable and capable of being cleaned and sterilised.	SR9
NFR2	The printer should be modular to support future upgrades.	SR15
NFR3	The printer must be affordable and within the allocated SU final year project budget.	SR11
NFR4	The design, development, and use of the bioprinter must comply with Stellenbosch University's health and safety regulations.	SR10, SR11
NFR5	The system should be user-friendly and simple to operate.	SR8
NFR6	The bioprinter should support multiple types of hydrogel.	SR5
NFR7	The system must include a documented procedures to calibrate the bioprinter to desired hydrogels and clean after use	SR9, SR12

The performance requirements, documented in Table 3.4, quantify how well the system must meet its functional requirements. The target levels for each performance requirement was chosen to match the performance capabilities of the current state of the art for bioprinters (such as the Bio X) as far as reasonably possible.

Table 3.4: Performance Requirements

No.	**Description**	**Target**	**Range**	**Unit**	**Related FR**
PR1	Extrusion speed[1]	5	1 - 10	mm/s	FR2
PR2	Positional accuracy along XY axes[2]	100	1 - 500	μm	FR1
PR3	Positional accuracy along Z axis[2]	50	1 - 100	μm	FR1
PR4	Maximum travel speed along XY axes[1]	10	0 - 50	mm/s	FR1
PR5	Minimum bed temperature[3]	4	< 20	°C	FR4
PR6	Maximum nozzle temperature[4]	85	$80 <$	°C	FR4
PR7	Nozzle temperature control tolerance	$\pm$ 1	$\pm$ 0.5 $-$ 2	°C	FR4
PR8	Extrusion force control tolerance	$\pm$ 5	$\pm$ 2 $-$ 10	%	FR2, FR7

[1] Recommended print speed of 0-10 mm/s for the BIO X (CELLINK, 2023).
[2] The Bio X has a 1 µm XY/Z positional accuracy (CELLINK, 2025).
[3] This matches the Bio X's minimum bed temperature capability of 4 $°C$ (CELLINK, 2025).
[4] The target matches the maximum capability of the Bio X's pneumatic print heads (CELLINK, 2025).

3.3 Extruder Design

The rest of this chapter will cover the design of each of the bioprinter's subsystems, using a concept generation and selection process, followed by the final design. Since the system's two primary functions are to extrude gel and position the extrusion nozzle in 3D space, these functions serve as the basis for dividing the system into its respective subsystems (extruder and gantry). This section focuses on the extruder design, starting with the development of generated concepts.

3.3.1 Concept Generation

Concept 1

The first concept, shown in Figure 3.1, illustrates a pump-based extruder design in which a peristaltic pump is used to drive the continuous flow of non-gelled bioink from a fluid reservoir. The reservoir could take the form of either a permanent or intermediate temporary storage for the gel ink (configured to create the correct handling conditions) and not necessarily in the beaker-shaped form shown in the figure.

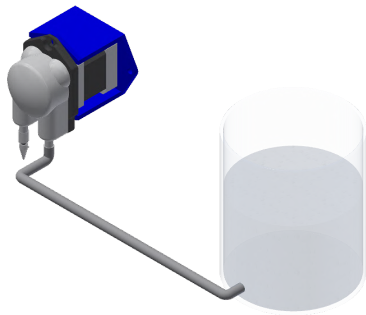


Figure 3.1: Concept 1 Pump-Based Extruder Design

Concept 2

Figure 3.2 shows the second concept is an assembly, in which a stepper motor drives two lead screws (through a system of gears) to push a plunger further into a syringe. The pre-existing syringe is easily removable by either unbolting the flanged tube in which the syringe sits or by sliding the syringe out vertically when the platform that pushes the plunger is in its uppermost position. The inclusion of gears is used to allow the plunger to be driven with a greater force magnitude for extruding high-viscosity gels through small-diameter nozzles.

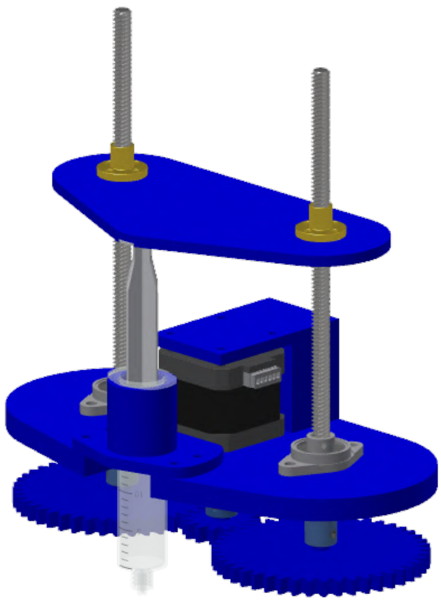


Figure 3.2: Concept 2 Syringe-Based Extruder Design

Concept 3

The final concept, in Figure 3.3, shows another syringe-based extruder similar to the second concept. This concept deviates from the previous one with regards to its frame and actuation style. This concept uses a syringe with a threaded nozzle to accept screw-in component additions (for added modularity). The extruder uses a belt-and-pulley drive-train to increase the force of the lead screw on the plunger and also uses a shaft to guide the plunger-driving platform.

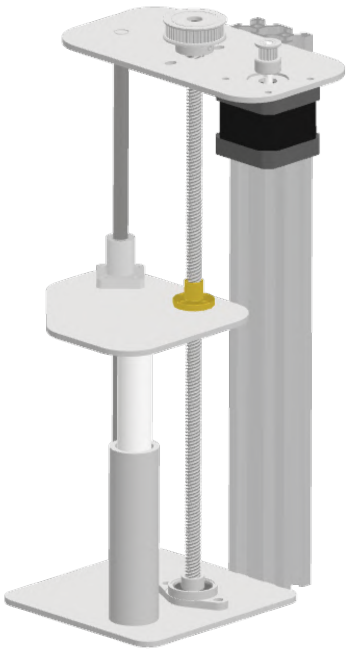


Figure 3.3: Concept 3 Custom Syringe-Based Extruder Design

3.3.2 Concept Evaluation and Selection

To objectively evaluate the three generated concepts and identify the one that best satisfies the project's stakeholder requirements, a Pugh chart was developed. This method compares each concept against multiple weighted criteria, each derived from a specific stakeholder requirement. Concept 2 was selected as the reference datum, and the other concepts were rated relative to it on a scale from -2 to $+2$, with positive values indicating favourable performance and negative values indicating less favourable performance. Table 3.5 presents the results of the Pugh chart evaluation.

The Pugh chart indicates that Concept 3 provides the most favourable balance across the weighted criteria. Its lead screw and drive-train increases flow rate accuracy and extrusion force. Although scoring similarly to Concept 2 in most categories, its addition of belt tensioning means greater extrusion accuracy due to reduced backlash. These factors enable the extruder to handle highly viscous gels with repeatable performance while maintaining sterilisability.

Table 3.5: Concept Evaluation (Pugh Chart)

Criterion		Concept 1		Concept 2		Concept 3	
Description	**Weight**	**Rating**	**Motivation**	**Datum**	**Motivation**	**Rating**	**Motivation**
Maximum flow rate (SR1)	0.07	+2	Higher throughput	0	Limited by lead screw and drive-train	0	Limited by lead screw and drive-train
Maximum extrusion force (SR1)	0.13	-2	Head losses due to longer fluid path	0	High due to drive-train	0	High due to drive-train
Ink storage volume (SR1)	0.05	+2	Large reservoir and continuous pump action	0	Limited by syringe volume	0	Limited by syringe volume
Flow rate accuracy (SR3)	0.19	-1	Inaccurate at low speeds	0	Accurate due to lead screw and gear reduction	0	Accurate due to lead screw and drive-train
Repeatable performance (SR6)	0.15	-1	Tube ageing	0	Variable backlash from printed gears	+2	Low backlash from tensioning
Sterilisability and cleanability (SR9)	0.13	-2	Hard to clean long path	0	Removable syringe and straight path	0	Removable syringe and straight path
Affordability (SR11)	0.15	-2	Requires a peristaltic pump	0	Standard parts	0	Standard parts
Sum of plus ratings		4		0		2	
Sum of minus ratings		-8		0		0	
Overall total		-4		0		2	
Weighted total		-0.92		0		0.30	

3.3.3 Final Extruder Design

With Concept 3 selected as the most suitable option, the concept is refined into a final extruder design that is ready for assembly. In this subsection, the extruder is specified in detail, with calculations provided to quantify its design requirements (stepper torque, pulley ratios, and plunger dimensions) and to determine its theoretical performance characteristics such as flow rate resolution. These demonstrate that the final design meets the stakeholder requirements.

Extrusion Force Calculations

To calculate the needed stepper torque and pulley ratio, it is first necessary to describe the system characteristics and requirements. The syringe should hold roughly 25 mL and therefore has an inner diameter of 20 mm and an 80 mm length. The nozzle's threaded shank provides attachment for an E3D hot-end assembly that can accommodate different nozzle diameters. To be conservative and to allow printing with high accuracy, the design assumes the ability to extrude through a minimum nozzle diameter of 0.2 mm with an approximate length of 20 mm. To ensure compatibility with multiple hydrogels, the extruder is specified to handle gels with a maximum viscosity of 1 Pa·s to match GelMA's viscosity above 30 °C.

These assumptions are restated in Appendix A.1, which shows, using Poiseuille's Law, that extrusion under these edge-case conditions requires the plunger to deliver a maximum pressure of 181.3 kPa. The calculations further indicate that, to achieve this pressure over the cylinder's cross-sectional area, the lead screw must apply approximately 60 N of force on the plunger. With a safety factor of 5, 300 N corresponds to a torque requirement of 0.638 N · m for the TR8×8 lead screw. To achieve this, a NEMA17 stepper motor with a maximum holding torque of 0.28 N · m is used in combination with a 3:1 GT2 pulley reduction (48T:16T), increasing the maximum torque at the lead screw to 0.84 N · m.

Flow Rate Resolution

The minimum flow rate resolution without microstepping can be determined based on the relationship between the angle of the motor shaft, lead screw pitch, and the cross-sectional area inside the syringe. Since the system uses a pulley ratio (PR) of 3:1, the lead is 8 mm, and the motor is adjusted by 1.8° per full step, the positional accuracy with which the plunger can move is:

$$\begin{aligned}\Delta x_{\text{Full Step}} &= \frac{\text{Lead}}{\text{PR}} \times \frac{\theta_{\text{Full Step}}}{360^\circ} \\ &= \frac{8.00}{3} \times \frac{1.8^\circ}{360^\circ} \\ &= 0.0133 \text{ mm/step.}\end{aligned}$$

To convert this to a volumetric resolution, this value must be multiplied by the area of the syringe's cross-section through which the plunger slides:

$$\Delta V_{\text{Full Step}} = \pi \left(10 \times 10^{-3}\right)^2 (1.33 \times 10^{-5}) = 4.19 \times 10^{-9} \text{ m}^3\text{/step} = \boxed{4.19\ \mu\text{L/step}}.$$

Therefore, the extruder has a minimum theoretical extrusion resolution of 4.19 μL for a full step that drives the stepper. This resolution can be further refined through microstepping.

Buckling Analysis

The syringe plunger is made of PVC, and the minimum diameter of the plunger stem must be determined to avoid buckling under the extrusion-force edge case of 300 N. According to Callister and Rethwisch (2018), the modulus of elasticity of PVC ranges from 2.41 GPa to 4.14 GPa. Therefore, taking the modulus of elasticity as 2.41 GPa and considering that the plunger stem length is 80 mm (to allow full range of motion in the syringe), the minimum diameter can be determined with the Euler buckling formula and the area moment of inertia for a circular cross-section:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}, \quad \text{and} \quad I = \frac{\pi d^4}{64},$$

$$\iff d = \left(\frac{64P(KL)^2}{\pi^3 E} \right)^{\frac{1}{4}}.$$

For this derived formula for the stem's minimum diameter, it is further noted that the design includes parts that constrain the displacement at each end of the plunger (the syringe, and a circular hole into which a load cell is mounted and the plunger end slides). These parts constrain displacement but do not fix the plunger's angle at either end, so Euler's effective-length factor K is assumed to be 1 (pin-to-pin connection). Therefore, the minimum diameter is:

$$\begin{aligned} d &= \left(\frac{64 \times 300 \times (1 \times 0.080)^2}{\pi^3 \times 2.41 \times 10^9} \right)^{\frac{1}{4}}, \\ d &= 6.37 \text{ mm.} \end{aligned}$$

As an additional safety measure, the plunger stem is specified as 18 mm, which provides a large margin against buckling. This choice also reduces machining waste on the lathe, since the plunger head must be 20 mm in diameter to sit within the syringe tube.

Thermal Expansion Calculations

To allow the extruder to achieve its required maximum nozzle temperature of 85 °C (PR7), an E3D heater block is screwed onto the syringe's threaded nozzle and a standard 1.0 mm 3D printer nozzle is screwed into the heater block. However, the ability to elevate the syringe's temperature results in the thermal expansion of some of its components. PVC has a greater coefficient of thermal expansion than aluminium, which means that the plunger requires a minimum tolerance between the plunger head's outer diameter and syringe barrel's inner diameter must be determined to prevent the heated plunger from binding inside the syringe. According to Callister and Rethwisch (2018), the linear coefficients of thermal expansion for 6061 aluminium alloy and PVC are 23.6×10^{-6} (α_{Al}) and up to $180{\times}10^{-6}\,\mathrm{K}^{-1}$ (α_{PVC}). Therefore, the minimal diametric clearance must be (assuming $T_{\mathrm{room}} = 20°\mathrm{C}$):

$$\Delta d_{\mathrm{tol}} = (\alpha_{\mathrm{PVC}} - \alpha_{\mathrm{Al}})\, d\, \Delta T = (180 - 23.6)(1 \times 10^{-6})(20)(85 - 20) = \boxed{0.203 \text{ mm}}.$$

Viton O-rings are used to seal the plunger, especially when the extruder is operated at lower temperatures (increasing the clearance). Viton is selected as the O-ring material due to its high temperature resistance.

Final Design

The final extruder design is shown below in Figure 3.4.

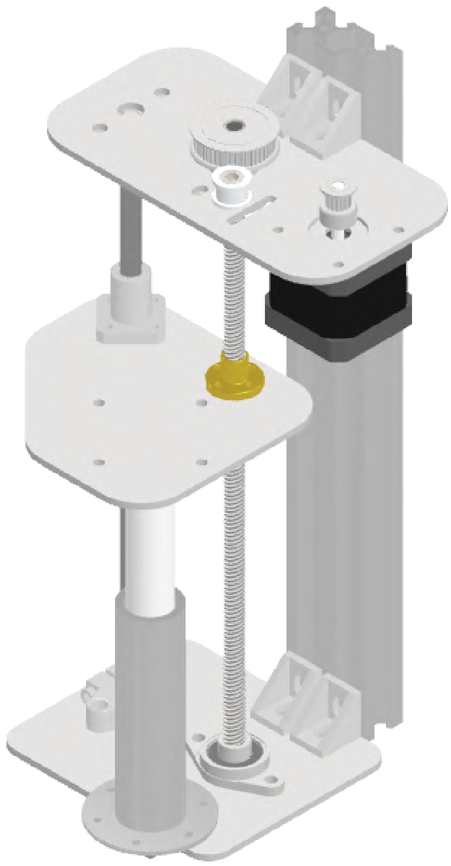


Figure 3.4: Final Syringe-Based Extruder Design

This figure shows the refined version of Concept 3. It adds brackets to mount the platforms to the 2020 aluminium extrusion, includes a slot-and-idler-based belt tensioning mechanism, separates the syringe from the lower platform into two parts mounted together via the syringe's flange. An E3D heater assembly is also mounted to the syringe's threaded nozzle in order to raise the temperature of the syringe. To assist readers in visualising the intricate features of the syringe design (such as O-ring grooves), a labelled exploded mechanical assembly drawing of the syringe assembly is included in Appendix B.

3.4 Gantry Design

The design of a gantry enables the positioning of the extruder, allowing the system to function as a 3D printer. This section follows the same structure as the extruder design in which concepts are developed and evaluated, a single concept is selected and refined into a final design.

3.4.1 Concept Generation

Concept 1

The first concept (Figure 3.5) uses a CoreXY system with two steppers driving the XY belts and a third stepper that lifts the bed via a lead screw. The frame is made from aluminium extrusions, with linear rails guiding all horizontal motion. The Z-axis bed is guided by two shafts and bearings and the bed is comprises of a removable copper plate (to aid in manual process like curing or cleaning).

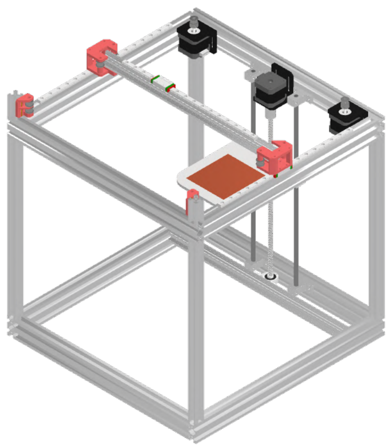


Figure 3.5: Concept 1 CoreXY Design

Concept 2

In Figure 3.6, each axis is driven by its own lead screw(s) (all turned by NEMA 17 stepper motors), with wheel carriages guiding motion along aluminium extrusions. The bed is supported by two linear shafts with bearings and also has a removable copper plate.

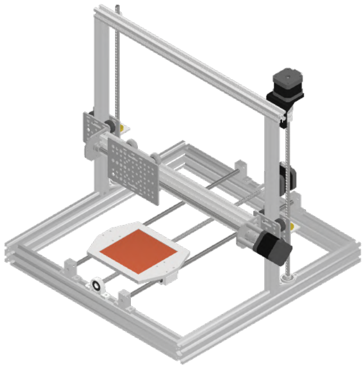


Figure 3.6: Concept 2 Bed-Slinger Design with only Lead Screws

Concept 3

The final design, in Figure 3.7, uses two lead screws (synchronized by a belt), driven by one stepper, to lift the Z-axis beam. The extruder carriage is guided along the Y-axis v-slot aluminium beam and driven by a belt, while another belt drives the bed in the X-direction along two rods (like in Concept 2). The bed is a fixed aluminium plate to enable the addition of a permanent cooling mechanism (Peltier).

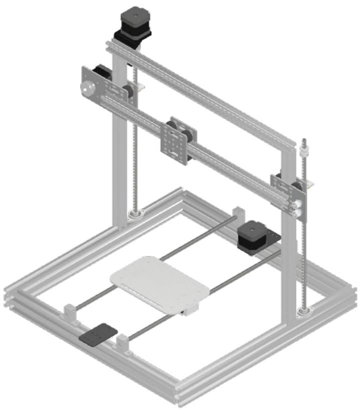


Figure 3.7: Concept 3 Bed-Slinger Design with Lead Screws and Belts

3.4.2 Concept Evaluation and Selection

The concepts are evaluated, as was done previously, in Table 3.6, which compares each concept against a set of weighted criteria (based on the project's stakeholder requirements). These criteria include factors such as positional accuracy, speed, affordability, and sterilisability. Each factor was compared against Concept 2 (the datum) using a relative score of up to ±2 (as with the extruder evaluation). The resulting scores were then totalled according to each criterion's respective weight.

Table 3.6: Concept Evaluation (Pugh Chart)

Criterion		**Concept 1**		**Concept 2**		**Concept 3**	
Description	**Weight**	**Rating**	**Motivation**	**Datum**	**Motivation**	**Rating**	**Motivation**
Maximum XY speed (SR1)	0.12	+1	Uses belts	0	Limited by lead screw	+1	Belts can achieve higher speeds
XY positional accuracy (SR2)	0.24	-1	Belts are less accurate	0	High accuracy of lead screws	-1	Belts are less accurate
Z positional (layer height) accuracy (SR2)	0.24	-1	Uses lead screws but is susceptible to bending	0	Accurate due to lead screws	0	Accurate due to lead screws
Sterilisability and cleanability (SR9)	0.20	0	Removable copper bed	0	Removable copper bed	-1	Fixed bed
Affordability (SR11)	0.20	-1	Requires more parts (Idlers and tensioners)	0	Uses more expensive parts (lead screws compared to belts)	+2	Simplest build with cost effective, standard parts
Sum of plus ratings		1		0		3	
Sum of minus ratings		-3		0		-2	
Overall total		-2		0		1	
Weighted total		-0.56		0		0.08	

The results of Table 3.6's evaluation reveal that Concept 3 is the most suitable option of the three generated concepts, achieving the highest overall weighted score and therefore best satisfies the system's stakeholder requirements. While it is more difficult to clean/sterilise the bed after prints due to it being fixed to the linear bearings, it provides a base to mount a Peltier module for bed cooling which would have been impractical if the bed was removable due to wiring and heat sinks. It also performs favourably in terms of speed and affordability, due to its use of standard GT2 belts that are often used in 3D printing. It can also achieve high layer height accuracy due to its use of synchronised lead screws.

3.4.3 Final Gantry Design

Finally, Concept 3, needs to be refined and developed into a final gantry design that is ready for procurement and assembly.

Z Axis Stepper Requirements and Positional Accuracy

To determine the stepper torque requirements, the mass driven by the synchronised lead screws must be approximated. This can be estimated using Inventor Professional's iProperties feature, which revealed that the total volume of the portion of the final design (extruder, horizontal beam, motors, and carriages) being lifted in the Z-direction is $688\,658.130\,\text{mm}^3$. By assuming the average density of this assembly is $2.7\times10^{-6}\,\text{kg/mm}^3$ (the density of aluminium), the mass to be moved by the lead screws is approximated as $1.86\,\text{kg}$.

Appendix A.2, using the same lead screw properties as A.1, shows that this requires a minimum stepper torque of $0.0388\,\text{N·m}$. The final design uses a Creality 42-40 Stepper Motor that is capable of a holding torque of up to $0.4\,\text{N·m}$. The lead screw torque therefore has a safety factor of roughly 10.

Finally, the positional accuracy of the Z-control must be characterised. The chosen stepper has a step angle of 1.8° and the TR8x8 lead screws have a lead of 8 mm. Therefore the positional resolution is:

$$\begin{aligned}\Delta z_{\text{Full Step}} &= \text{Lead} \times \frac{\theta_{\text{Full Step}}}{360^\circ} \\ &= 8.00 \times \frac{1.8^\circ}{360^\circ} \\ &= 0.04 \text{ mm/step.}\end{aligned}$$

This theoretical accuracy outperforms the 50 µm requirement established in PR3. It should be noted that an even higher resolution can be achieved with microstepping.

X and Y Axis Stepper Requirements and Positional Accuracy

Autodesk Inventor Professional can be used to estimate the final extruder's total volume of $379\,353.531\,\text{mm}^3$ and the printer bed with its linear bearings, heat sink, Peltier and cold plate (shown in the final design later in Figure 3.8) has a total volume of $145\,105.084\,\text{mm}^3$. As done previously, by multiplying these volumes by the density of aluminium ($2.7\times10^{-6}\,\text{kg/mm}^3$), yielding two respective mass approximations of $m_x = 0.39\,\text{kg}$ and $m_y = 1.02\,\text{kg}$. By further assuming the carriages' friction coefficient to be 0.2 and the maximum horizontal acceleration to be $a = 5\,\mathrm{m/s}^2$, the required stepper torque can be determined for each axis, as shown in Appendix A.3. The X-axis uses a GT2 pulley with 20 teeth and the calculations show that the stepper requires 0.0173 $N \cdot m$ of torque and the Y-axis stepper (which uses a 36 teeth pulley) needs 0.08137 $N \cdot m$. Therefore, the chosen NEMA 17 steppers have a holding torque up to 0.28 N·m for the X-axis and 0.4 N·m for the Y-axis.

With these described specifications, the positional accuracy in horizontal plane can be determined. Considering the use of pulleys in the X and Y directions, the resolution can determined by multiplying the motors step angle with the radius of each pulley ($r_x = 6.365$ and $r_y = 11.46$):

$$\Delta x = \theta_{\text{Full Step}} \times r_x = 1.8^\circ \frac{\pi}{180^\circ} \times 6.365 = 0.200\,\mathrm{mm},$$
$$\Delta y = \theta_{\text{Full Step}} \times r_y = 1.8^\circ \frac{\pi}{180^\circ} \times 11.46 = 0.360\,\mathrm{mm}.$$

These fall short of the target set in PR2, but both fall within the acceptable range. However, they can be configured to meet the target by using the drivers to enable microstepping of at least 1/4.

Final Gantry

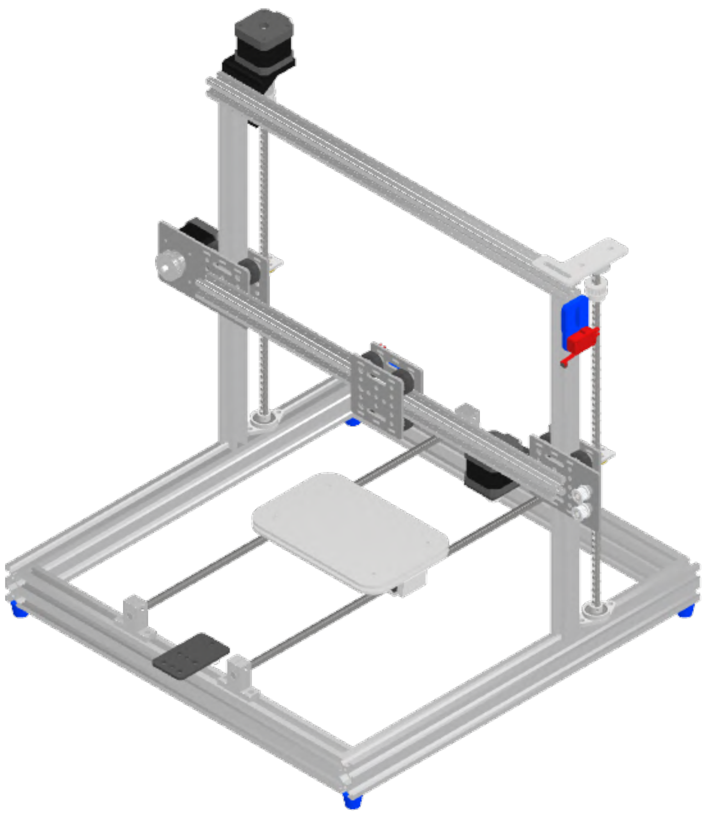


Figure 3.8: Final Bed-Slinger Gantry Design

The final gantry design is shown above in Figure 3.8. It is the refined design, based on Concept 3, and incorporates a Peltier cooler mounted to the bed between two plates (cold side on top and heat sink at the bottom). The figure also shows the system's limit switches for homing.

3.5 Final Design

The complete bioprinter design, shown in Figure 3.9, shows the extruder and gantry subsystems, integrated into a single mechanical system. The extruder is designed to use a custom syringe, heater block, and load cell to deposit viscous gels at a controlled flow rate and temperature, while the gantry accurately positions it in 3D space to enable it to print three dimensional scaffolds. This final design concludes the mechanical design of the project and forms the basis for the embedded systems design and integration to enable the printer to function as required and undergo experimental validation. To provide a more detailed view of the bioprinter's design, the mechanical drawings of all custom parts, as well as the gantry and extruder assembly drawings, are available in the project's GitHub repository's `Hardware` folder (Daniel, 2025).

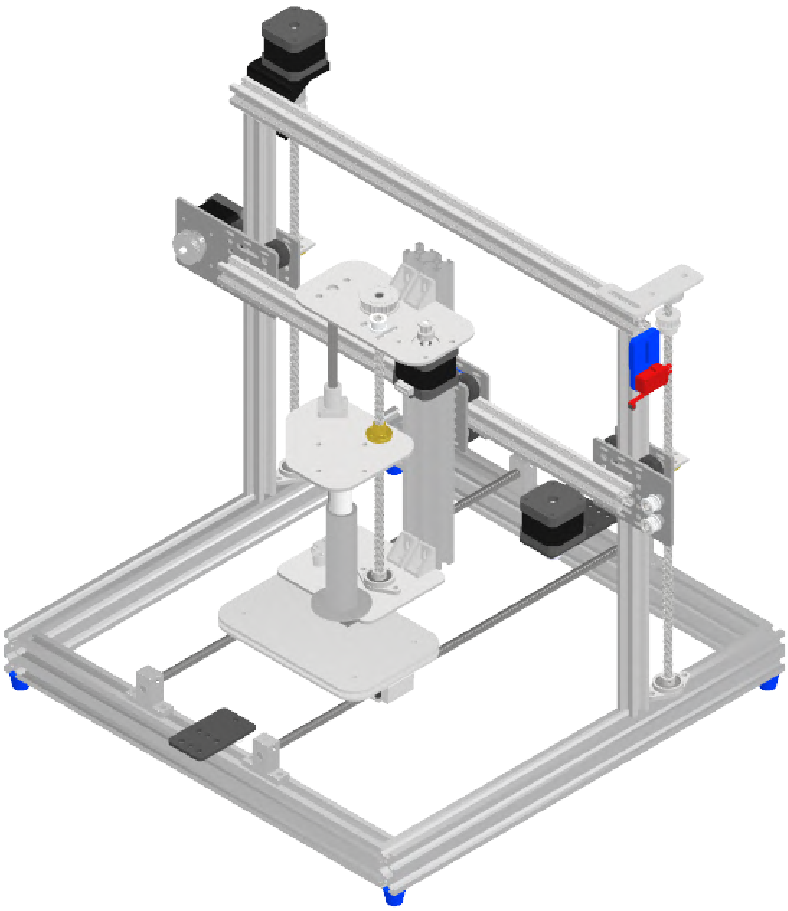


Figure 3.9: Final Bioprinter Design

Chapter 4

Embedded System Design

This chapter details the design and implementation of the embedded control system for the hydrogel bioprinter. The embedded system is responsible for coordinating the mechatronic system's core functions, including stepper movement, memory management, bed and nozzle temperature control, and UV curing. The electronic hardware design takes the form of a custom PCB featuring an STM32F407VGT6 microcontroller, which executes the developed firmware to allow the printer to function. While more affordable alternatives exist, the STM32F407VGT6 was selected based on prior experience, thereby reducing development time and the learning curve.

The embedded system was designed to meet the following requirements, developed for the electronic design in order to allow the whole system to meet the stakeholder requirements from chapter 3:

- The system must safely supply regulated 3.3 V to the logic subsystems from a 12 V DC power supply that plugs into a standard wall socket.
- The system must be able to drive up to five stepper motors through A4988 driver breakout modules.
- The system must be able to receive real-time sensor data, including extrusion force from a load cell and temperature readings from thermocouples.
- The device must implement switching control for a heater cartridge, Peltier module, UV curing light, and cooling fan.
- The system is required to receive print task via a microSD card.
- The printer must implement feedback control loops for extrusion pressure and temperature regulation.
- The system must provide headers for unused controller pins to support future upgrades.

The final assembled hydrogel bioprinter, including the implemented embedded system, is shown below in Figure 4.1:

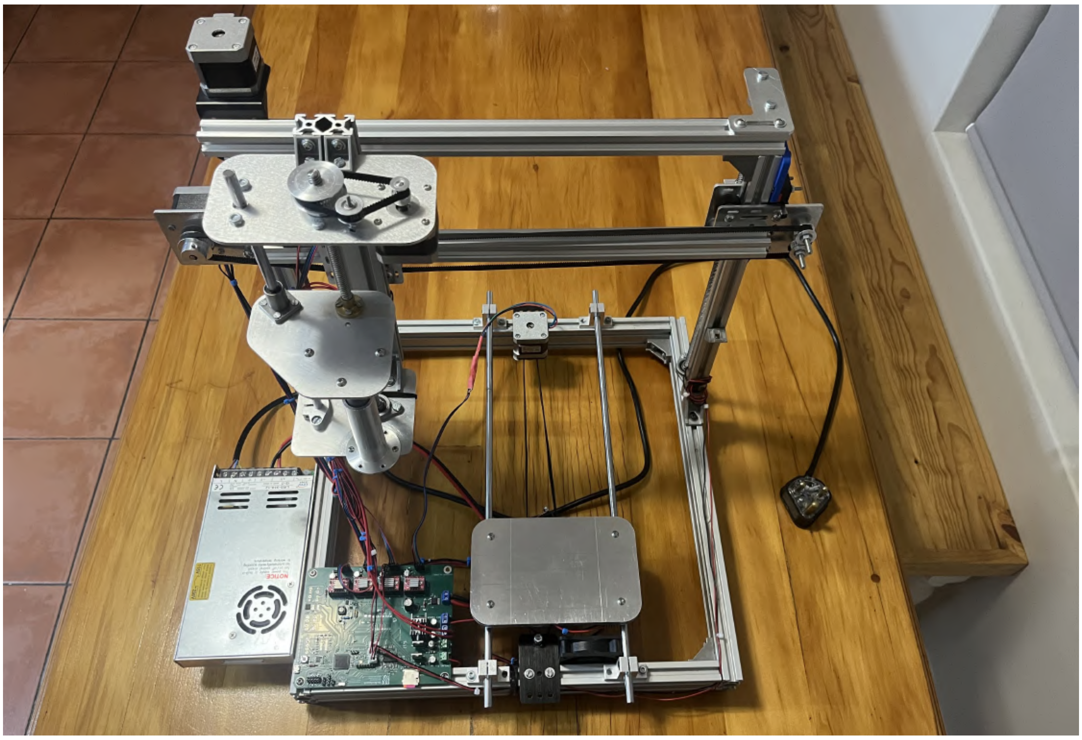


Figure 4.1: The Final Assembled Hydrogel Bioprinter

4.1 System Description

The embedded system as a whole enables the printer's automated operation. It is powered by a 12 V PSU that directly supplies power to the stepper drivers, Peltier module, UV curing light, heater cartridge, and cooling fan. The 12 V is also converted into a regulated 3.3 V source via a buck converter to power the logic circuits. The STM32 controller reads print instructions from a microSD card so that it can then control the stepper motors (through drivers) using an ISR. During stepper operation, system information is transmitted to a laptop via a USB connection for external monitoring. External peripherals (Peltier, heater cartridge, cooling fan, and curing light) are connected to the 12 V source and switched between their ON and OFF states using N-channel power MOSFETs.

This description of the system is visually represented in the block diagram below in Figure 4.2. The black arrows represent the logic flow, the red arrows show the 12 V power distribution, and the green arrow shows how the 3.3 V power source is distributed.

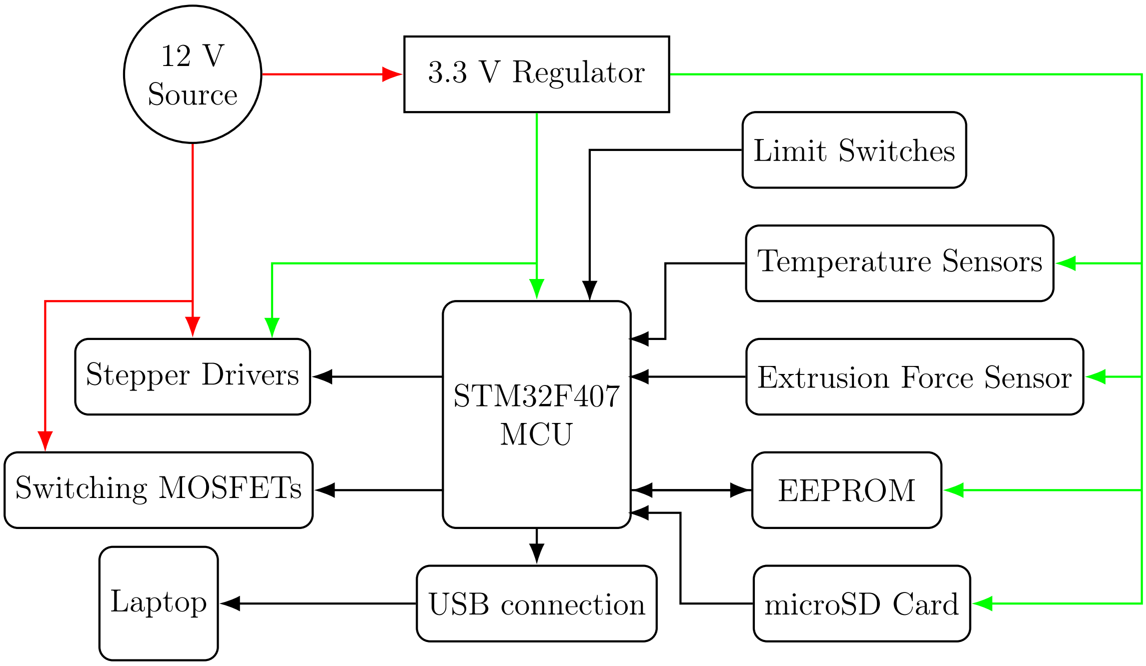


Figure 4.2: System Block Diagram of the Bioprinter's Embedded System

4.2 Hardware Design and Implementation

The hardware design, as previously stated, took place in the form of a custom PCB. The PCB was designed using KiCad and manufactured by JLCPCB. The electronic system can be separated into various subsystems, including the power supply, stepper drivers, limit switches, memory handling circuit, load cell sensing circuit, temperature reading circuit, and the N-channel MOSFET switching circuit. The timing of each subsystem is coordinated by a high-speed crystal oscillator. A schematic overview of the PCB is shown in Figure C.4 in Appendix C.

4.2.1 Power Supply

The power supply subsystem, shown in Figure C.5, incorporates both circuit protection and voltage conversion through a buck converter. The printer is powered by a 350 W LRS PSU, which allows direct connection to a standard wall socket and converts the municipal 230 V_{RMS} supply into a regulated 12 V DC output. This output is delivered to the PCB via screw terminals.

In the event of incorrect wiring, the system makes use of reverse polarity protection using a high-side P-channel MOSFET. This ensures that, if the ground and 12 V connection are swapped, the source output is 0 V instead of -12 V. This method was chosen over a simple series diode to eliminate forward voltage drop and preserve the full 12 V for the stepper drivers and switching peripherals.

The entire circuit board also uses bulk and decoupling capacitors to filter out a wide range of frequencies to produce stable voltage levels for both 12 V and 3.3 V components. The 12 V source also makes use of TVS diodes (SMAJ16A) positioned near inductive loads, such as the buck converter, stepper drivers, and cooling fan port, to suppress transient voltage spikes.

This subsystem also has the role of producing a stable 3.3 V source for the logic circuitry. To do this, the PCB uses an AP3211 buck converter, following the typical 12 V to 3.3 V application circuit by Diodes Incorporated (2018).

4.2.2 Power Debug LEDs

The electronic hardware incorporates debug LEDs to show the presence of specific voltage levels. A green LED ($V_f \approx 2\,\text{V}$) is connected to the buck converter's 3.3 V output to confirm that the system is connected to power and that the input voltage is successfully converted to 3.3 V. A second debug LED, a blue LED ($V_f \approx 3\,\text{V}$), is connected to the USB port's 5 V pin to show when the board is connected to a laptop via USB. Both LEDs require current limiting resistors, which were calculated to limit the current to 10 mA as follows:

$$R_{3.3\text{V}} = \frac{3.3\text{ V} - 2.0\text{ V}}{10\text{ mA}} = 130\ \Omega \qquad \text{and} \qquad R_{5\text{V}} = \frac{5.0\text{ V} - 3.0\text{ V}}{10\text{ mA}} = 200\ \Omega.$$

For an added safety margin, 220 Ω resistors were used for both LEDs.

4.2.3 Stepper Driver Interface

To interface with the stepper motors, the PCB employs five A4988 driver modules, as shown in Figure C.6. Three drivers control the motors of the 3-axis gantry, one drives the extruder stepper, and one spare driver is included to allow future upgrades, such as the addition of a second extruder. The A4988 drivers come in the form of Pololu breakout modules that plug into female headers on the PCB (visible in the 3D model in Appendix B.9). The advantage of using socketed breakout drivers is that they can be easily replaced in the event of damage. The headers follow the StepStick form factor which means the A4988 drivers can be replaced with more advanced drivers such as the TMC2209. While each driver requires a dedicated GPIO pin for its STEP and DIR inputs, to conserve controller pins, the $\overline{En}$, $\overline{SLEEP}$, $\overline{RESET}$, and the microstepping setting pins (M0, M1, and M2) are shared across all five drivers.

To limit the current, the potentiometer on each stepper driver needs to be adjusted to reach a desired reference voltage which can be obtained using the following formula:

$$I_{\text{limit}} = \frac{V_{\text{ref}}}{8R_{\text{sense}}} \qquad \Longleftrightarrow \qquad V_{\text{ref}} = 8R_{\text{sense}} \cdot I_{\text{limit}}.$$

By looking at the drivers and checking the stepper motor specifications, it can be concluded that:

- $R_{\text{sense}} = 0.10\ \Omega$
- $I_{\text{limit}} = 1.0\ \text{A}$

$$V_{\text{ref}} = 8 \cdot 0.10 \cdot 1.0 = 0.8\ \text{A}.$$

4.2.4 Limit Switches

Figure C.4 shows that the circuit board provides four limit switch connection points. Only three are utilised to allow for the homing of each gantry axis. The COM pin of each limit switch is connected to ground and the NO pin is connected to a dedicated GPIO input pin (PC0-PC2) that is pulled high so that when each axis reaches the end of its range of motion, the switch gets triggered and is pulled low.

4.2.5 Memory and Connectivity Interfaces

The board employs various means of handling memory and connecting with external devices, both for reading and transmitting data as well as for programming and debugging. This functions are achieved through the inclusion of an EEPROM, a microSD port, a USB port, and an SWD interface (Figure C.4).

The controller is connected to an M24C64 EEPROM, which provides a means of storing non-volatile information via an I^2C connection, ensuring that data can be retained even after the printer's power source is disconnected. In addition to this, the circuit board incorporates a microSD card port for reading print task files through the MCU's SPI1 channel. A USB port is also included, allowing the board to connect to a laptop and transmit live data—such as temperature and extrusion force readings for external monitoring and analysis. Because the USB interface requires a highly accurate clock speed (48 MHz) for reliable data transmission, the PCB includes a 16 MHz external crystal oscillator to provide a precise reference for the phase-locked loop (STMicroelectronics, 2024).

For programming and debugging, the board includes a 2×5 male header array. Each pin in the header is connected to the relevant system points, including the 3.3 V supply, ground, and the required STM32 pins (SWDIO, SWCLK, SWO, and NRST). During development, these SWD pins were connected to the corresponding pins of an STM32F407G-DISC1 development board to enable programming and debugging via its integrated ST-Link interface.

To improve modularity and support future upgrades, the board includes expansion headers that connect to unused MCU pins (Figure C.5). These pins connect to spare GPIO, ADC, USART, I^2C, and SPI channels, enabling potential additions such as external displays.

4.2.6 Load Cell Circuit

Figure C.7 shows the circuits sensor system which includes extrusion force sensing using a load cell amplifier. The amplifier circuit design was based on the SparkFun open-source HX711 load cell amplifier breakout module (Wende and Seidle, 2019). This is powered by the board's 3.3 V supply and connected to two controller GPIO pins (a clock pin and an input pin). The board has four relevant screw terminal connection points for an external load cell (RED, BLACK, WHITE, and GREEN).

The 50 kg load cell used to measure the extrusion force is configured as a half Wheatstone bridge (containing two $1\,\mathrm{k}\Omega$ resistors). To form a complete Wheatstone bridge, two additional $1\,\mathrm{k}\Omega$ resistors are required, as shown in the full load cell–to–PCB set-up in Figure 4.3.

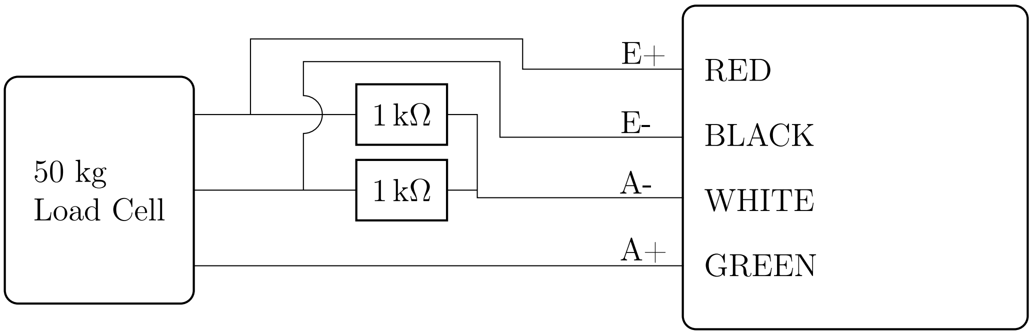


Figure 4.3: Load Cell Setup

4.2.7 Thermocouple Circuit

The temperature sensing circuit is shown in Figure C.7. This uses a MAX6675 K-type thermocouple, which is connected to the STM32's SPI2 pins (PB12-PB14). The amplifier receives the analogue thermocouple input, converts the signal into a digital value, and outputs this to the controller via the Serial Peripheral Interface.

To allow the controller to read multiple thermocouples using a single amplifier, an analogue multiplexer (TMUX1208) is used. The MUX cycles through eight two-pin thermocouple screw terminals, where the T+ line connects to the amplifier input and the T− line is tied to ground. Channel selection is controlled by the MUX's enable pin (EN) and three select pins (A0, A1, A2). The binary combination of these pins determines which of the eight thermocouple channels is connected to the amplifier input at a given time.

4.2.8 FET Switching Circuit

Figure C.8 shows the control circuit responsible for switching the system's various 12 V peripherals. These 12 V peripherals (each connected via two-pin screw terminals), include a Peltier cooling module, an E3D heater cartridge, a cooling fan, and a UV curing light. For safety reasons and due to the use of a non-curable surrogate gel in the experimental procedure (described in Chapter 5), a blue led was used in place of the UV light to demonstrate functionality.

The peripherals are switched using N-Channel power MOSFETs (IRLZ44N), which are capable of handling high currents required by components such as the Peltier module (maximum current 7 A). The transistors are configured as low-side switches, with their source terminals tied to ground, drains connected to the negative side of the loads, and gates driven by STM32 GPIO outputs. If fully implemented, a UV light would be kept in a fixed position and flood the print bed with an adjustable intensity, which would be set by a duty cycle read from the microSD card during print commands.

A potential issue with this arrangement is that the transistor gates act as capacitive loads. When GPIO outputs transitions from high to low, the stored charge from the gate capacitance discharges into the MCU pin as a temporary transient discharge current. Since each GPIO pin is only rated for 25 mA (STMicroelectronics, 2024), this risks damaging the controller. To mitigate this, small series resistors are placed at each gate to limit the initial discharge current. In addition, an octal bus transceiver (SN74HC245DWR) is used as a protective buffer stage between the MCU and the MOSFET gates. This configuration not only shields the microcontroller from harmful transient currents but also improves signal integrity when driving multiple high-capacitance gates.

4.2.9 PCB Design

The embedded hardware was implemented on a custom PCB designed in KiCad and fabricated by JLCPCB. Its design required a number of engineering decisions regarding trace widths, layer count, and overall routing strategy.

The PCB is a two-layer board, with the bottom copper layer serving primarily as a ground plane and the top layer carrying power and signal routes. The complete board layout is shown in Figures C.1 to C.3. In areas supplying the 12 V peripherals, such as the switching MOSFETs and stepper drivers, the top copper includes solid planes to provide sufficient current capacity while minimizing temperature rise.

Signal routing on the top layer generally uses 0.3 mm traces, while 3.3 V power lines were widened to 0.5 mm. The 0.5 mm trace width is justified through the empirically derived IPC-2221 current-temperature relationship, to estimate a minimum width (W) of a 1 oz/ft^2 thick copper trace to conduct the maximum buck-converter output current of 1.5 A with a 5 °C temperature rise:

$$W = \frac{1}{t_{\text{cu}}} \left(\frac{I}{k(\Delta T)^{0.44}} \right)^{\frac{1}{0.725}} = \frac{1}{1.378} \left(\frac{1.5}{0.048(5)^{0.44}} \right)^{\frac{1}{0.725}} = 16.9 \text{ mil} = 0.43 \text{ mm.}$$

Communication traces connecting the SD card, EEPROM, USB interface, MAX6675 thermocouple amplifier, and HX711 load cell amplifier to the STM32 microcontroller were deliberately kept short and closely routed to the MCU, with spacing between traces maximised (as far as reasonably possible) to preserve signal integrity. Similar care was taken around the buck converter circuitry to minimize interference arising from its high-frequency operation.

4.3 Software Design and Implementation

The firmware governs the 3D printer's operation. After initialising its peripherals, the system homes the gantry to its reference position and zeros the coordinate axes. It then enters an interrupt-driven routine that handles all stepper motion while the main loop cycles through system tasks such as SD card reading, USB transmission, and MOSFET duty cycle updates. The system's general software flow is visually represented in Figure 4.4.

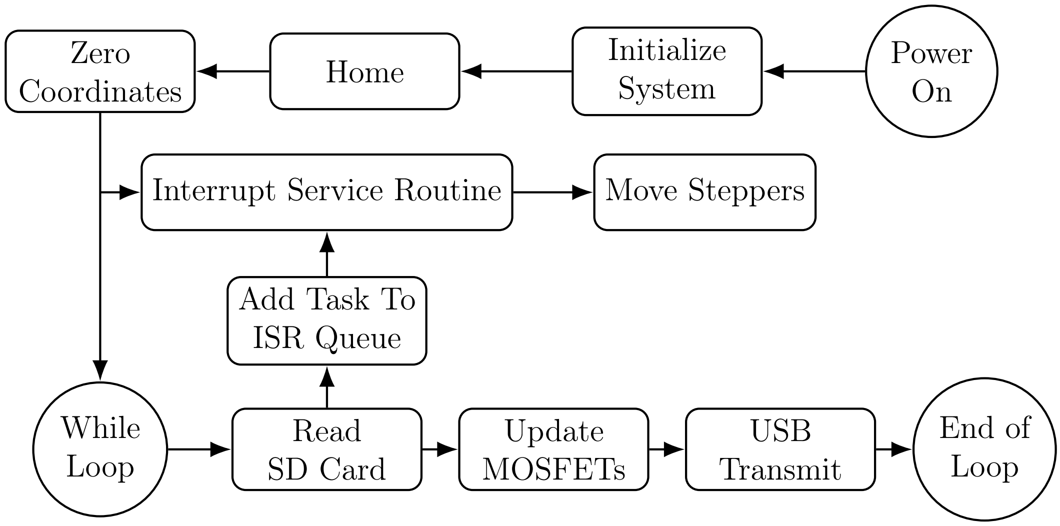


Figure 4.4: Bioprinter's Generalised Software Flow Diagram

4.3.1 Homing

Upon startup, after all system peripherals have been initialised, the printer performs a homing sequence in which the gantry moves each of its three axes towards known reference positions (0, 0, 0). This procedure establishes the origin of the its coordinate system, ensuring that its absolute extruder position for a given coordinate remains consistent between power cycles.

The A4988 drivers are enabled (EN low, RESET and SLEEP high) and configured for 1/16 microstepping (M0,M1, and M2 high). Each driver's DIR pin is set so that each axis moves towards the corresponding limit switch. Each normally-open limit switch is connected to a dedicated GPIO input pin that is pulled high internally. When the limit switch closes, the input pin is connected to ground and goes low.

Step timing is handled by TIM5, which runs at 1 MHz (PSC = 83), giving a 1 µs counter tick. Each step pulse is held high for 5 µs and then low for an axis-specific duration so that each axis moves at approximately 16 mm/s. During homing, only one axis moves at a time until its limit switch GPIO input is no longer high, at which point that axis stops and the next axis begins. The axes are homed in the order of Z, Y, and lastly X.

4.3.2 Stepper Logic

After the gantry is homed, the printer's motion system is scheduled through an interrupt service routine (ISR). The advantage of timing the stepper movements with scheduled interrupts, rather than triggering each step from within the program's `while(1)` loop, is that step pulses are consistently generated on time and remain unaffected by lower-priority tasks that could introduce delays. Although the firmware may not currently be performing computationally intensive operations that make this approach strictly necessary, it provides future-proofing by ensuring reliable motion control even as more time-demanding features (G-code parsing) are added.

Once the system is homed, the enable pin is set low, and the sleep and reset pins are set high as instructed by Allegro MicroSystems, LLC (2014). The microstep selection pins are all driven high to configure the driver for 1/16 microstepping mode, ensuring maximum positional accuracy for the belt-driven axes. The DIR pins for each driver are pre-defined in the code as high or low to define each axis' positive direction. Before the ISR starts executing the commands from the SD card, the stepper coordinates are set to zero with the command `Stepper_ResetPosition(x0, y0, z0, e0)`.

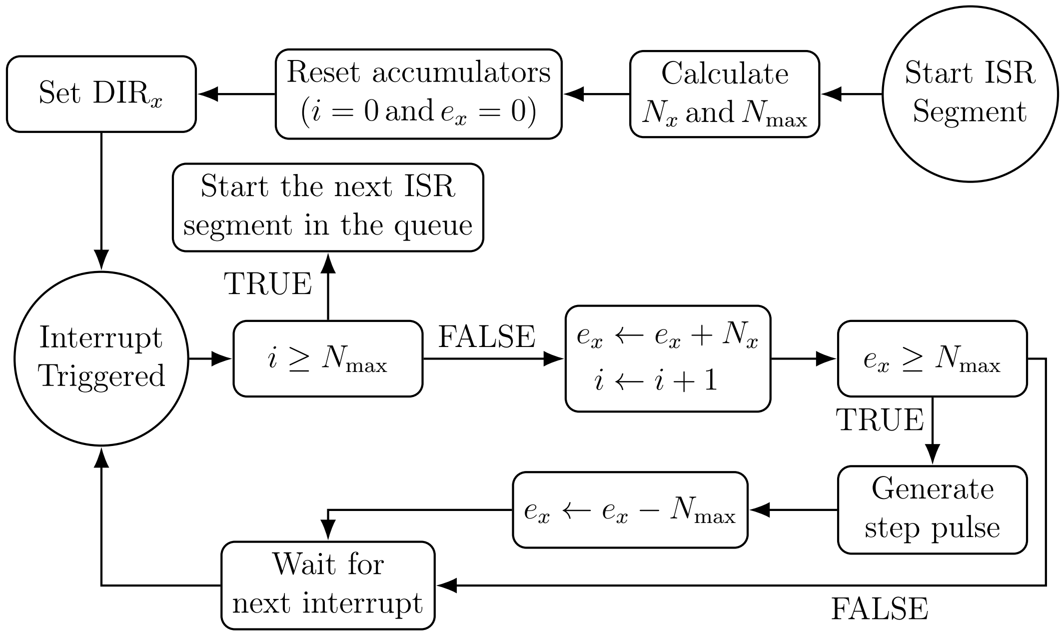


Figure 4.5: Software Diagram of an ISR Segment on the X-axis

The ISR, driven by TIM1, implements motion commands executed from a circular queue. The commands use the format of x position, y position, z position, speed, extrusion rate, and a boolean input that determines whether the command is implemented according to a relative (1) or absolute (0) coordinate system. From this, the ISR operates with the following steps (shown in Figure 4.5):

- The relative distance each axis must move is calculated (i.e. $\Delta x = x_{target} - x_{current}$).
- Using the steps per millimeter for each stepper, the required number of steps is determined (i.e. $N_x = |\Delta x \times \text{STEPS\_PER\_MM}|$).
- The direction level (DIR) is determined for each driver by considering the sign of each axis' position change and the positive direction level defined in the code.
- To determine the total execution time of each command, the travel length is calculated ($L = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$), and the total time is then obtained by dividing this length by the gantry speed ($T = \frac{L}{v}$).

- To ensure that all axes arrive simultaneously, the maximum step count is calculated as $N_{max} = \max(N_x, N_y, N_z, N_e)$.
- From this, the ISR runs at a frequency of $f_{\text{step}} = \frac{N_{\text{max}}}{T}$, and the interrupt period is the reciprocal.
- At each interrupt period, each axis increments its own Bresenham accumulator to determine whether a pulse must be generated for any of the drivers. For example, the x axis accumulator, e_x, increments by N_x during each interrupt, but a step pulse is only generated for the x axis once $e_x \geq N_{\text{max}}$, in which case the accumulator will also be reduced by N_{max}.

4.3.3 USB Communication

The STM32 controller uses a USB Full-Speed CDC virtual COM port on PA11 and PA12 using the STM32 USB Device library. Upon startup, the USB system is initialised and a 1 Hz task transmits a message to communicate system information (such as temperature) via `CDC_Transmit_FS()`. The transmission is sent in the following format:

```
setNozTemp=30.00C, NozTemp=26.12C, Load=5.71 kg \n
```

A laptop can connect to the PCB via a USB cable and view the stream using any serial terminal (such as Termite), or transform the data in MATLAB during the printer's operation. The RX callback is set up but is not utilized. This introduces the opportunity for later firmware upgrades to allow the possibility of external commands being sent from a laptop for additional control.

4.3.4 SD Card Reading

The printer reads motion commands from an SD card using the FatFs middleware over SPI communication (SPI1). Once FatFs is initialized and the file system is mounted, a command file, `PrintTask.csv`, is opened read with the `f_open()` function.

During printer operation, the next commands (if the ISR queue has space) are read in the program's `while(1)` loop every 50 ms (20 Hz)task. The format of each command line in the CSV file is:

$$X,\ Y,\ Z,\ \text{Speed},\ \text{ExtrusionRate},\ \text{Mode},\ \text{NozzleTemp},$$

where `Mode` selects absolute (`0`) or relative (`1`) moves. Read command lines are queued into the ISR via `Stepper_QueueAbs()` or `Stepper_QueueRel()`.

4.3.5 Switching Logic

The printer switches its 12 V peripherals (nozzle heater, cooling fan, UV LED, and Peltier module) by controlling the logic levels of the gate terminals of their N-channel MOSFETs via the microcontroller. Upon startup, the Peltier gate (PB4) is driven high, keeping the cooler continuously on. The remaining three loads, are controlled via PWM through the STM32's timer channels:

- Nozzle heater: `TIM3_CH2` on `PB5`
- Fan: `TIM4_CH1` on `PB6`
- UV LED: `TIM4_CH2` on `PB7`

With PSC = 0 and ARR = 4199, each pulse width modulation channel operates at 20 kHz:

$$f_{\mathrm{PWM}} = \frac{f_{\mathrm{timer}}}{(\mathrm{PSC}+1)(\mathrm{ARR}+1)} = \frac{84\ \mathrm{MHz}}{(0+1)(4199+1)} = 20\ \mathrm{kHz}.$$

Duty cycles (0-100 %) are updated in the `while(1)` loop every 50 ms by adjusting the channel compare register (CCR):

$$\mathrm{CCR} = \frac{\text{Duty Cycle}}{100}\,\mathrm{ARR}.$$

The blue LED undergoes a linear brightness modulation over a 10-second periodic interval (to show curing intensity adjustment capabilities) and the fan is set to operate at full intensity ($D = 100\%$).

4.4 Measurements and Results

To verify nozzle heating and bed cooling, a FLIR ONE Pro camera captured the temperature distribution of both components (100% duty cycle) after 10 minutes of operation, as shown in Figures 4.6 and 4.7. Due to an unavailability of hand-held thermocouple's in the M&M Workshop during testing, a CASTALWARE digital thermometer was used to determine less accurate settling temperatures. By resting the end of the thermometer on the bed's centre, a minimum of 13 °C was recorded, and placing the probe in the E3D heater block's thermistor hole indicated a maximum of 230 °C.

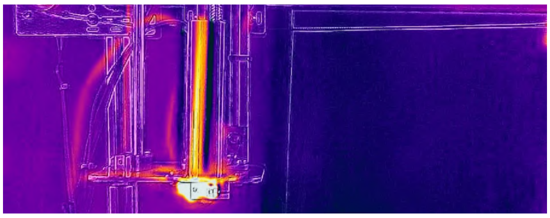


Figure 4.6: Nozzle temperature distribution after 10 minutes of being active (100% duty cycle)

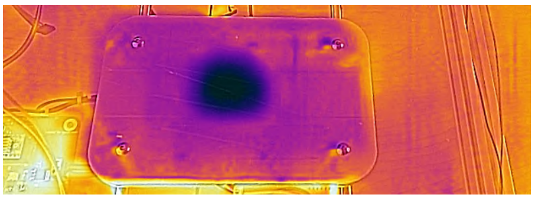


Figure 4.7: Bed temperature distribution after 10 minutes of being active

Chapter 5

Experimental Design

The purpose of this chapter is to provide an experimental procedure that allows the bioprinter to be calibrated for a specific hydrogel and to ensure user-safety and reliable operation by providing handling and cleaning instructions.

The printer's purpose is to serve as a research tool for Stellenbosch University's Physiology Department to characterise unknown bioinks and to fabricate hydrogel scaffolds to model and simulate cell behaviour in biomedical research. To ensure the hydrogel extrusion system is integrated effectively into this research environment, the chapter establishes protocols for adjusting printing parameters to find the optimal settings for different gels, and details procedures to handle the bioprinter and to clean it after use to maintain hygiene and prevent contamination between experiments which ensures cell viability in cases where the printer is used to print cell-seeded structures. The procedure was demonstrated on a surrogate gel for future reference. These guidelines are intended to ensure that the bioprinter can be used consistently, safely, and effectively throughout its utilisation phase.

The calibration procedure involves applying a series of different experiments with the bioprinter and its bioink to optimise its print parameters within different categories of print operations (printing straight, curved, multiple adjacent, and stacked lines).

5.1 Materials and Methods

5.1.1 Materials

The experimental set-up requires the following equipment and materials:

- Custom 3D printer with syringe-based extruder
- HTS Silicone Grease
- Allen Key
- Laptop
- microSD card with microSD card adapter
- Bioink solution

- Camera
- Calliper

For the surrogate gel, the following equipment and materials are required:

- Store-bought Agar powder
- Kettle
- Measurement jug
- Measurement spoons (1 tsp, 1/2 tsp)

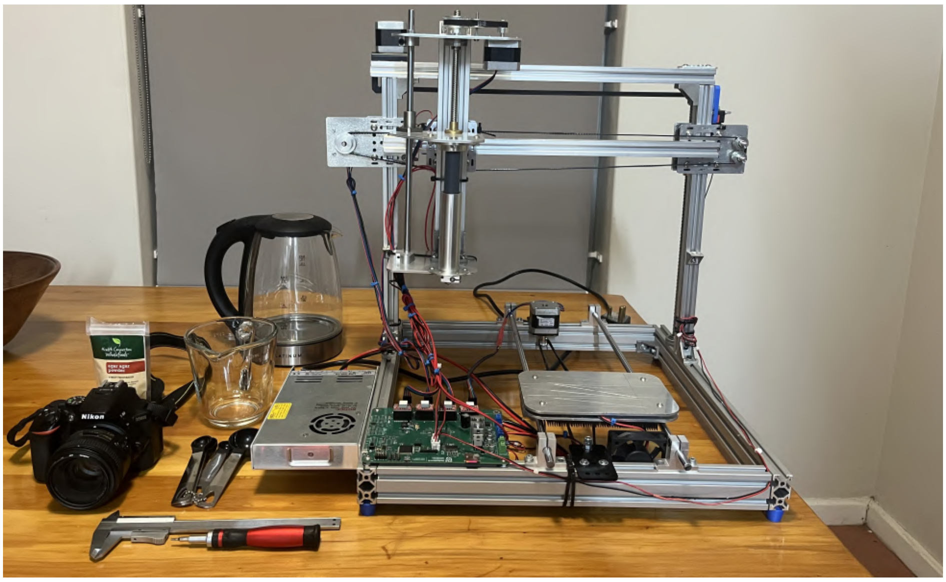


Figure 5.1: Experimental Set-up

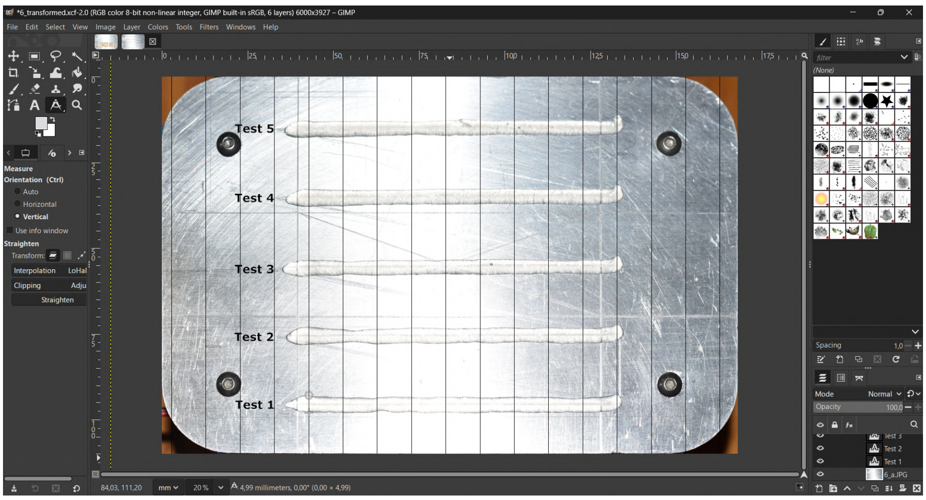


Figure 5.2: GIMP Set-up

The bioprinter is comprised of an extruder and a gantry. The extruder uses a lead screw to extrude the gel out of a custom syringe onto a cooled aluminium bed. The extruder position relative to the printer frame, is varied across 3D space by a gantry that uses belts to adjust the extruder position in the horizontal plain and uses a lead screw to adjust the nozzle height.

The bioprinter's print instructions are loaded via a microSD card which is inserted into and read by the bioprinter to perform each calibration trial. The laptop is also used to read transmitted USB information which is read in Termite.

After each print iteration, a photo is taken of the print bed and its dimensions are analysed using GIMP (Figure 5.2) to assess the print quality. This experimental set-up is shown in Figure 5.1.

5.1.2 Equipment Specifications

The experimental equipment must allow multiple strands of hydrogel to be extruded onto a printing surface where the line's width can be measured at points along the line without distorting the line's dimensions during the measurement process.

Gel Specifications

The gel specifications depend on the research needs of the end-user, with each hydrogel type and concentration coming with its own rheological properties and storage and handling protocols.

This experimental procedure, however, was applied to a surrogate gel that is an agar-water mixture. Store-bought Agar-agar powder is dissolved in water at 4.5% w/v. Agar powder was used as it is easy to prepare and gels more rapidly when cooled compared to other store-bought alternatives such as gelatin. Agar is a suitable surrogate for GelMA as they are both thermally reversible and shear-thinning hydrogels whose viscosities that decrease with increasing temperature and solidify upon cooling, making agar a low-cost substitute for testing.

Bioprinter Specifications

- **Extruder** – This consists of a 25 ml syringe with a nozzle diameter of 1.0 mm. The nozzle heater block is capable of reaching a maximum temperature of 230 °C.

- **Gantry** – The gantry is used to position the extruder in 3D space. Two NEMA 17 stepper motors drive belts that control the 2-axis motion in the horizontal plane. A lead screw (also driven by a NEMA 17) with an 8 mm lead is used to control the extruder's vertical position.

- **Bed** – The gel is printed on a 110 mm $\times$ 168 mm aluminium bed. The bed is capable of reaching a minimum temperature of 13 °C.

Camera Specifications

Print sample photos are taken using a Nikon D5600 (DSLR) with an AF-S NIKKOR 50 mm f/1.8G SWM aspherical ϕ58 lens. The camera operates in aperture priority mode with flash enabled, an aperture of f/4.5, and an image resolution of 6000 × 4000 pixels. Photographs are taken by hand under ambient lighting conditions.

Software

- **GIMP** – This is used to distort the image to remove the angled perspective effect that varies between hand operated camera shots and measure the dimensions of the print strands for each print trial.
- **MATLAB** – The measured data from the printing trials are stored manually and processed in MATLAB using the `Trial_Processing.m` file in the project's GitHub repository's `Experiments` folder (Daniel, 2025).

5.1.3 Methods

The variables identified, relating to the calibration procedure, are shown below in Table 5.1:

Table 5.1: Experimental Variables

Variable Type	**Variables**
Independent Variables	Extrusion rate, print speed, bed-nozzle height, print arc radius, line spacing
Dependent Variables	Layer width, layer height, print stability
Controlled Variables	Gel concentration, print line length, nozzle diameter
Extraneous Variables	Ambient temperature, vibrations, gel rheological properties

During the analysis of results, the mean line width, the standard deviation of the line width (each in mm) along with the overall quality of each print trial is analysed and compared against the varied printing parameters: extrusion rate (mm/s), printing speed (mm/s), bed-nozzle height (mm), print circle radius (mm), horizontal line spacing (mm), and nozzle height increment (mm). The standard deviations serve as measures of print stability (by indicating consistent width), where high deviations indicate lower print stability and quality. Because the rate at which the ink gets extruded depends on the nozzle diameter, extrusion rate (in this chapter only) is defined as the speed (in mm/s) at which the plunger moves through the syringe.

The controlled variables in the experiment are the nozzle diameter and gel concentration. The nozzle diameter influences the width of deposited strands and is therefore kept constant across all trials. The gel concentration affects the fluid behaviour of the solution and is held constant to minimise variations in flow properties between print tasks.

Ambient temperature and environmental vibrations, including those from the printer itself, can also influence outcomes. However, these are considered extraneous variables and are not within the scope of control in this experiment. Since the printer is intended to characterise unknown gels, the gel's rheological behaviour, even under constant concentration, is considered as an extraneous variable.

Gel preparation

The following steps detail how to prepare the agar surrogate gel.

1. Fill the kettle with water and heat water until boiling (100 °C).

2. From the kettle, pour the boiling water into a clean measurement jug until the bottom of the meniscus reaches the 100 mL mark on the measuring jug.

3. Using the measuring spoons, add 1.5 teaspoons (4.5 g) of agar powder to the measurement jug.

4. Mix the solution until homogeneous, and continue stirring every 10 minutes until an hour has passed since the solution was first stirred.

5. After the first stir, complete steps 1 to 4 of the experimental set-up procedure.

Experimental Set-up Procedure

Before the calibration, the a set-up procedure must first be conducted.

1. Plug the printer's three-prong power plug into a standard power socket. The PCB indicator light should turn green, the gantry will home, and the bed cooling system will activate.

2. Using the laptop, load the print settings and print tasks (according to the needs of the specific experimental trial that is being conducted) onto the microSD card.

3. Using the Allen Key, unbolt the syringe from the extruder assembly and unscrew the syringe from the hot-end assembly.

4. Check and ensure syringe is clean and press the plunger until it is fully depressed to the bottom of the syringe tube.

5. Once the gel solution is ready for use, submerge the tip of the syringe into the substance and pull the plunger until it reaches the end of the syringe or a sufficient volume of the solution is extracted.

6. Once the printer fluid has been loaded, wipe the syringe sections that were submerged during extraction with a cloth.

7. Screw the hot-end assembly back on to the threaded syringe tip and mount the flanged syringe back onto the extruder platform using the bolts and Allen Key (make sure at least two bolts are fastened along the flange mounting holes.

8. Insert the card into the bioprinter circuit board's microSD port, and press the reset button to begin printing.

9. Do not touch the printer during operation and keep the area around moving parts and nip points clear of loose items.
10. Once the print trial is done, use the camera to take a overhead photo of the print bed.
11. Analyse the results in GIMP.
12. Perform clean-up procedure.

Cleaning Procedure

To clean the bioprinter after use the following steps are taken:

- Make sure the printer is disconnected from its power source, and wait for the circuit board's green light to completely turn off.
- With a cloth, remove any prints from the printer bed and wipe the platform.
- Use the Allen key to remove the syringe from the extruder platform.
- Unscrew both the nozzle and syringe from the heater block.
- Remove the plunger and pour any residual unsolidified liquid into a sink.
- Rinse both the syringe and nozzle with warm water.
- Fill the syringe with water and use the plunger to push it out to ensure there is no clogging.
- Leave the syringe, nozzle, and plunger to dry.
- Once dry apply a fresh coat of HTS silicone grease to the plunger head before reassembling the extruder.

GIMP Processing

Once the photos are taken, they must be processed and analysed in GIMP.

- Using the transformed tools scale the image to 6000x3928 to match the bed dimension's ratios.
- Then use the perspective tool and drag the bed edges to precisely align the demarcated rectangle.
- Go to `Scale Image` in the `Image` tab and set the X and Y pixels per mm to 35.7 and the image size should precisely match the bed dimensions of 168x110 mm.
- Show a grid using `View` and then using `Configure Grid` in the `Image` tab, create horizontal spacings of 10 mm to ensure line widths are equally spaced, create vertical spacings of 0.1 mm to ensure 0.1 mm measurement accuracy, and offset the grid horizontally by 153.6 mm to ensure the vertical grid-lines align with the etched bed centre.

- Using the measuring tool, measure the required line widths and manually store the results in the `Trial_Processing.m` file in the project's GitHub repository's `Experiments` folder (Daniel, 2025).

Nozzle Height Test

Once the setup is complete, the calibrations can begin. 6 different printing phases are conducted with each phase using its own PrintTask.csv file. These CSV files, along with the MATLAB scripts that generated them, can also be found in the `Experiments` folder of the project repository (Daniel, 2025). Each print task is repeated multiple times, and the trials exhibiting the fewest instances of temporary clogging or entrapped air bubbles (identified by random gaps along deposited strands) are included in the appendices (the rest are shown in the GitHub repository for transparency). This approach ensure the independent variable is the only factor influencing the dependant variables. A description of the print plan of the nozzle height test can be seen below.

1. The first experiment is to determine the optimal vertical nozzle height to print straight lines. 5 adjacent 100 mm straight lines are printed on the bed with a gantry speed of 20 mm/s and extrusion rate of 0.2 mm/s with 5 different nozzle offsets as in the proposed test matrix in Table D.1. This task is repeated 5 times.
2. The width of the 100 mm lines are measured at 11 equi-spaced points and the mean and standard deviation of these widths for each line are calculated.
3. The results are evaluated in the Discussion of Results section to determine the optimal height value, which is then used to print the lines in the next calibration phase.

Speed Test

1. The second experiment determines the optimal print speed and extrusion rate. 16 straight lines (100 mm long) are printed across two print tasks with varying extrusion rates for different print speeds according to Table D.2. This task is repeated 5 times.
2. The width of the 100 mm lines are measured at 11 equi-spaced points and the mean and standard deviation of these widths for each line are calculated.
3. The results are evaluated in the Discussion of Results section to determine the optimal height value, which is then used to print the lines in the next calibration phases.

Curved Arc Test

1. The third experiment determines the optimal centripetal acceleration used to print a curved line. It does so by printing 5 concentric circles with differing radii according to the proposed test matrix in Table D.3. The previously determined print settings are used including tangential print speed which is

held constant to ensure that the centripetal acceleration is differing due to the change in radius ($a_c = \frac{v^2}{r}$). This task is repeated 5 times.

2. The width of each circle's lines will be measured at 12 equi-spaced points (every 30°), and the mean and standard deviation of these widths for each circle are calculated, analysed, and evaluated in the Discussion of Results section.

Adjacent Line Test

1. The goal of the fourth experiment is to determine the optimal horizontal spacing with which the nozzle is offset by when printing horizontal rectangles. It does so by printing 6 sets of 4 adjacent 50 mm lines, with each set varying in how much the nozzle is horizontally offset (proposed offsets in Table D.4). The print settings determined from previous calibration stages are used and held constant between each rectangle speed. This task is repeated 5 times.
2. The results are process in GIMP and qualitatively analysed in the results section to determine the maximum line spacing trial that never forms gaps between adjacent strands.
3. The optimal line spacing is used to print the next calibration task.

Layer Stacking Test Procedure

1. The goal of the fifth experiment is to determine the optimal vertical nozzle height increase between successive scaffold layers. This is achieved by printing three 4 layer prisms with a target base of 30x30 mm, each with a different height increment (Table D.5). The previously determined optimal print settings are used and held constant across each cuboid. This is repeated 4 times.
2. An overhead photograph is still taken for this trial to be processed in GIMP and an additional angled photograph is captured to illustrate the 3D nature of the print.
3. The calliper's depth rod is used to determine the overall scaffold height by inserting it until it contacts the bed surface, then sliding the calliper body down until its end aligns with the top of the printed structure. The calliper is kept vertical throughout this process.
4. The results are analysed to identify the vertical offset that produces the best print quality.

Repeatability Test Procedure

1. To determine the repeatability of the printer with its optimised print settings, a sixth trial is conducted using the nozzle offset, extrusion rate, and print speed from Trials 1 and 2. Five 100 mm lines are printed on the bed. This is repeated three times.
2. The width of the 100 mm lines are measured at 11 equi-spaced points and the mean and standard deviation of these widths for each line are calculated.

3. The results are evaluated in the Discussion of Results section to assess the repeatability of the printer's performance.

5.2 Discussion of Results and Conclusion

Trial 1 - Nozzle Height Test

The processed image of the first printing trial is shown in Figure D.1, and the corresponding measurements obtained in GIMP and further processed in MATLAB are summarised in Table D.6.

The nozzle height that produced the line with the most consistent width and a printed length closest to the 100 mm target was 0.4 mm. Consequently, all subsequent print trials were performed with a first-layer nozzle offset of 0.4 mm.

The line printed at the smallest offset (0.2 mm) was noticeably shorter. This is likely because the minimal gap maximised the contact area between the nozzle and the gel, thereby increasing the adhesive forces at the nozzle tip. As a result, the initial portion of the extruded strand tended to follow the nozzle for a brief period before settling, yielding a shorter printed strand.

To assess whether the independent variable – the nozzle height – exhibited any meaningful correlation with the measured line properties, correlation coefficients were calculated in MATLAB. The correlation between nozzle height and line length, mean width, and width standard deviation was 0.131, -0.333, and -0.330. These weak correlations indicate that nozzle height did not have a strong influence on the printed line length, width, or variability. Since no statistical relationship was found between nozzle height and print quality, the 0.4 mm offset was selected as a reasonable baseline for the subsequent calibration phases.

Trial 2 - Speed Test

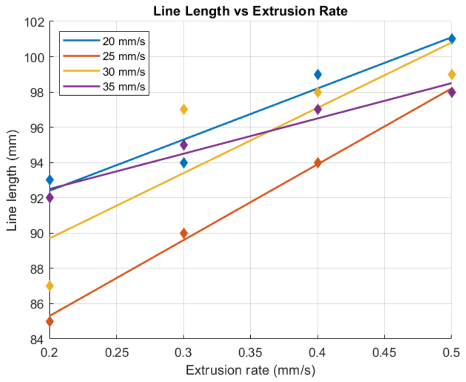
Figures D.2 and D.3 show the processed images of the speed printing trial, in which print speed and extrusion rate were varied to evaluate their combined influence on line quality. The corresponding results are tabulated in Table D.7, and the correlation analysis is shown in Table 5.2.

The plots in Figure 5.3 illustrate the relationships between the line characteristics and extrusion rate at each print speed. These figures show that increasing the extrusion rate produces a consistent rise in mean line width for all speeds, each exhibiting a clear linear relationship. This trend matches the expected behaviour, as a higher extrusion rate increases the volumetric flow of gel through the nozzle, thereby depositing more gel per unit time (when print speed is kept constant) and resulting in wider lines. This is further corroborated by the correlations in Table 5.2 which show a strong positive correlation ($r = 0.97$–1.00).

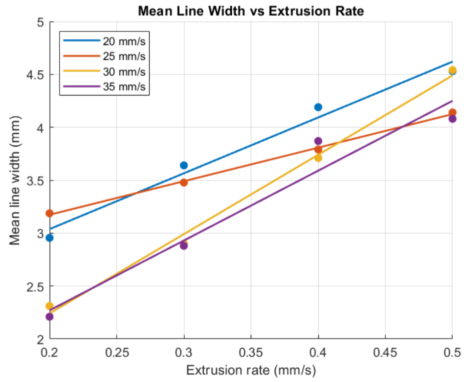
The plots and correlation analysis show a strong relationship between line length and extrusion rate ($r = 0.86$–1.00). Lines printed at higher extrusion rates were longer and more closely matched the 100 mm target length. This occurs because faster extrusion establishes the gel flow sooner, allowing the gel to contact and adhere to the bed closer to the intended start. At lower extrusion rates, the nozzle travels further before the delayed flow can achieve gel deposition, resulting in a shorter line.

The effect of extrusion rate on the line width variation differed between print speeds. At print speeds between 20 and 25 mm/s, no correlation was found ($r \approx 0$), indicating that increasing extrusion rate at low speeds does not affect print line width stability. When print speed increased to between 30 and 35 mm/s, the correlation became strongly positive, showing that higher extrusion rates at elevated speeds increased the likelihood of irregularities forming along the line.

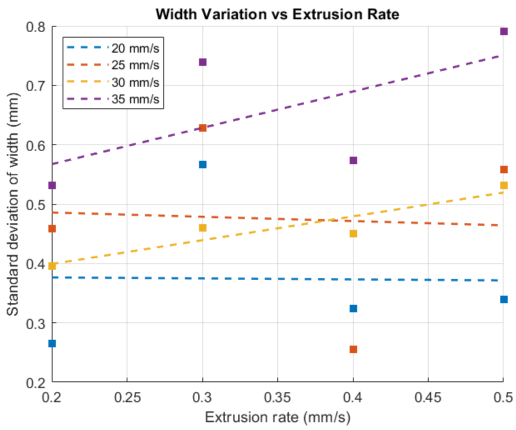
These results show that lines printed with lower print speeds and higher extrusion rates result in the most uniform and repeatable lines that more closely match the target line length. Based on this trend, a print speed of 20 mm/s and an extrusion rate of 0.4 mm/s were selected for the rest of the calibration process, as this setting yielded the most favourable combination of line width stability (with a mean of 4.2 mm) and length (99 mm).



(a) Length



(b) Mean Width



(c) Width Standard Deviation

Figure 5.3: Graphs showing the relationship between line characteristics and extrusion rate

Table 5.2: Correlation between line characteristics and extrusion rate at different print speeds.

Print Speed (mm/s)	Length	Mean Width	Std Dev
20	0.97	0.99	−0.04
25	1.00	1.00	−0.06
30	0.86	1.00	0.92
35	0.98	0.97	0.62

Trial 3 - Circle Test

The processed image of the bed with the printed circles is shown in Figure D.4 and each line's arc angle, mean width, and width variation are summarised in Table D.8.

The arc angles traced out by the printed circles ranged from 352° to the 360° target. The tabulated results show little to no correlation between the arc angle and circle radius. MATLAB analysis yielded a correlation coefficient of 0.205 between radius and arc angle, indicating a weak positive relationship. Although weak, this relationship is consistent with expected behaviour as circles with larger radii correspond to longer arc lengths for a given angular displacement. Consequently, any delay before gel deposition will subtend a smaller angle (resulting in larger deposited arc angles) on circles with a larger radius compared to smaller ones.

The correlation coefficient between the mean width and circle radius was -0.052, indicating no significant relationship. This outcome is expected as the extrusion rate and print speed were held constant at the values determined in the previous calibration phase. As a result, the total volume of gel deposited per unit length of nozzle travel remained consistent, yielding mean widths (4.5 - 4.7 mm) that closely matched the mean width from Test 3 of Trial 2 (4.2 mm), which had the same print speed and extrusion rate.

The correlation coefficient between radius and the standard deviation of the arc width was -0.518, indicating a moderate negative correlation. This suggests that the variability in strand width decreases as the radius increases which is expected as the centripetal acceleration of the printer nozzle decreases with larger radii, resulting in smoother motion.

These tests showed that the printer can accurately trace circular paths with minimal width variability. The circle with a radius of 5 mm exhibited the highest width variability, with a standard deviation of 1.16 mm. This suggests that circles with radii below 15 mm, should be printed at reduced tangential speeds (and proportionally lower extrusion rates to maintain line width) to limit centripetal acceleration to a maximum value of $a_c = \frac{(20\ \mathrm{mm/s})^2}{15\ mm} \approx 26.7\ \mathrm{mm/s}^2$ to ensure consistent line quality.

Trial 4 - Line Spacing Test

The results of the adjacent-line test are shown in Figure D.5. In this iteration, the line spacing between adjacent parallel lines was varied to determine the maximum horizontal offset that could be applied repeatedly without gaps forming between successive lines during any iteration of the print trial. This trial was repeated five times, with Figure D.5 illustrating one representative iteration.

Gimp was used to overlay a reference grid and to mark, in yellow, the intended start and end positions of the target rectangles defined in the input CSV file. The discrepancies between the yellow target lines and the actual printed rectangle edges were consistent for across all 6 tests. This consistency is expected, as the print speed and extrusion rate were held constant, and the initial deposition delay remained approximately the same for each batch. This explains why a small gap appears between the leftmost yellow line and the start of each rectangle, while the rightmost edge aligns closely with the intended endpoint, as each line was printed from left to right.

While the minimum line spacing that prevented gaps varied between iterations, only the first three test batches consistently produced fully connected lines across all five iterations. The fourth (4.5 mm) and fifth (5.0 mm) batches occasionally formed without gaps but were not repeatable. Therefore, a line spacing of 4.0 mm was selected for the next calibration stage, as it represented the largest spacing that reliably produced rectangles without visible gaps.

Trial 5 - Layer Stacking Test

A top and angled view of the stacked-layer print are shown in Figures D.6 and D.7. As before, the top view includes an overlaid reference grid with yellow squares indicating the target boundaries of the base of each four-layer cuboid. The heights of the cuboids were measured across four different print iterations using a calliper, and the results are shown below in Table 5.3.

Table 5.3: Measured cuboid heights for different nozzle offsets

Nozzle Offset (mm)	Height 1 (mm)	Height 2 (mm)	Height 3 (mm)	Height 4 (mm)	Mean (mm)
1.0	4.0	4.0	4.5	4.5	4.2
1.5	5.0	5.0	5.5	5.5	5.2
2.0	5.0	5.0	6.0	6.0	5.5

From Figure D.6, the effect of nozzle height increment on how well the printed bases matched the target boundaries can be observed. The first prism, printed with an offset of 1.0 mm between layers, matched the target boundaries the least, with its outer edges extending up to 5 mm beyond the yellow borders. The second and third batches (1.5 and 2.0 mm increments) aligned more closely with the yellow markings, with the third batch showing marginal improvement over the second.

The poor alignment of the first prism is expected, as its mean total height was 4.2 mm. This indicates that the structure's height increased faster than the nozzle, causing the nozzle to become partially submerged in previously deposited layers. The submerged nozzle distorted the base by displacing deposited gel outward as it moved, pushing the edges beyond the intended boundary. In contrast, batches 2 and 3 had average heights of approximately 5.2 and 5.5 mm, corresponding to an average layer height of about 1.3-1.4 mm. This suggests that the nozzle should increase in height by this amount to prevent submersion and the consequent distortion. Since batches 2 and 3 used higher increments, their bases were more accurate and undistorted.

The third batch differed from the previous ones in that the striations formed by the alternating lines on the top layer were more pronounced. This outcome is expected. Because each layer's true height is about 1.3-1.4 mm, a 2.0 mm nozzle increment means that the gap between the nozzle and the upper surface gradually increased during printing. This larger deposition height allowed the extruded gel to fall a distance before contact, creating rounder strands and a more visibly striated top surface.

It can therefore be concluded that a nozzle height increment of 1.5 mm provides the most balanced performance. This value closely matches the measured layer height of 1.3-1.4 mm, ensuring that the nozzle remains above the previous layer while maintaining sufficient proximity to the scaffold's upper surface to produce smooth, undistorted structures with minimal striation. This trial concludes the calibration phase of the experiment and validates that the bioprinter is capable of producing repeatable, stable, multi-layered gel scaffolds, thereby confirming the overall success of the project.

Trial 6 - Repeatability Test

An additional trial was conducted to characterise the repeatability of the system when printing gel using the optimal straight-line parameters identified in Trials 1 and 2. A vertical nozzle offset of 0.4 mm was used, with a print speed of 20 mm/s and an extrusion rate of 0.4 mm/s. Five 100 mm lines were printed on the bed plate under these identical conditions, as shown in Figure D.8.

The results are summarised in Table D.9. The standard deviation of the widths along each line ranged from 0.21 mm to 0.45 mm, indicating that the local line width varied by less than approximately 450 microns for the majority of the samples, indicating reasonably high stability of the extruded line for the soft gel material.

Further MATLAB processing revealed a mean line width of 4.52 mm with a 95% confidence interval of [4.35, 4.69] mm, and a mean line length of 98.4 mm with a 95% confidence interval of [97.7, 99.1] mm. These narrow confidence intervals demonstrate that the printer can repeatedly produce lines of consistent length and width when printing with a fixed set of tuned parameters.

Overall, this trial confirms that the printer operates with minimal random error, which further validates the reliability of the calibrated system for producing gel scaffolds.

Chapter 6

Conclusions

6.1 Future Work

While the printer achieved its primary goals of producing repeatable, stable extrusion of hydrogel to create 3D scaffolds, certain functional requirements were not met due to the project's broad scope and the time constraints imposed by hardware setbacks, leaving them for future implementation.

These outstanding functions include temperature and pressure sensing with feedback control, even though the electronic system was designed to support them. Implementing pressure sensing for closed-loop PID control to achieve highly accurate extrusion rates remains a key feature for future iterations of the project. Similarly, temperature sensing for both the bed and nozzle is required to be implemented in the future to enable closed loop thermal feedback for bed and nozzle thermal regulation, ensuring consistent printing conditions and improved repeatability.

6.2 Final Conclusions

The bioprinter developed in this project successfully demonstrated accurate and repeatable extrusion capable of producing multi-layered scaffolds through a low-cost and modular design. The integration of the mechanical, electronic, and firmware subsystms provided the printer with the ability to control motion and flow parameters, enabling consist and adjustable printing outcomes after calibration with the surrogate agar gel.

Despite delays and unimplemented control loops, the printing system achieved its primary objectives by serving as a calibration tool for characterising unknown bioinks and using them to create scaffolds, thereby validating the experimental design.

Overall, the project was successful and demonstrated the potential for developing low-cost bioprinter alternatives that can serve as research tools in the biomedical field. It provides a strong foundation for future students to improve the design, complete the outstanding feedback control, and expand the calibration procedure to further improve print quality and capability. Ultimately, this research lays the groundwork for Stellenbosch University to develop a fully automated, high-quality bioprinter that can be integrated into the Physiology Department's research environment.

Appendix A

Calculations

A.1 Syringe Calculations

Given:

$$\begin{aligned}\text{Inner diameter of barrel } (\mathrm{ID}_1) &= 20\,\mathrm{mm}\\ \text{Length of barrel } (\mathrm{L}_1) &= 80\,\mathrm{mm}\\ \text{Inner diameter of nozzle } (\mathrm{ID}_2) &= 0.2\,\mathrm{mm}\\ \text{Length of nozzle } (\mathrm{L}_2) &= 20\,\mathrm{mm}\end{aligned}$$

Cross-sectional areas:

$$\begin{aligned}A_1 &= \pi\left(10\times 10^{-3}\right)^2 \approx 314\times 10^{-6}\,\mathrm{m}^2\\ A_2 &= \pi\left(0.1\times 10^{-3}\right)^2 \approx 31.4\times 10^{-9}\,\mathrm{m}^2\end{aligned}$$

Max extrusion speed at nozzle:

$$v = 5\,\mathrm{mm{\cdot}s^{-1}} = 5\times 10^{-3}\,\mathrm{m{\cdot}s^{-1}}$$

Volumetric flow rate:

$$Q = v\cdot A_2 = (5\times 10^{-3})\cdot(31.4\times 10^{-9}) = 1.57\times 10^{-10}\,\mathrm{m^3{\cdot}s^{-1}}$$

Assumed viscosity:

$$\mu = 1\,\mathrm{Pa{\cdot}s}$$

Poiseuille's Law:

$$\Delta P = \frac{8\mu L Q}{\pi r^4}$$

Total pressure required:

$$\begin{aligned}P_{\mathrm{atm}} &= 101\,325\,\mathrm{Pa}\\ P_{\mathrm{Plunger}} &= \Delta P_{\mathrm{ID}_1} + \Delta P_{\mathrm{ID}_2} + P_{\mathrm{atm}}\\ &= \frac{8\cdot 1\cdot 0.08\cdot(1.57\times 10^{-10})}{\pi(0.01)^4} + \frac{8\cdot 1\cdot 0.02\cdot(1.57\times 10^{-10})}{\pi(0.0001)^4} + 101325\\ &= 181\,325\,\mathrm{Pa} = \boxed{181.3\,\mathrm{kPa}}\end{aligned}$$

Required force:

$$F = P \cdot A_1 = (181\,325) \cdot (314 \times 10^{-6}) = 56.94\,\mathrm{N}$$

Design force (with safety factor):

$$F_{\mathrm{SF}} = 300\,\mathrm{N}$$

Required torque for lead screw to produce axial force:

- Axial force: $F = 300\,\mathrm{N}$
- Coefficient of friction: $\mu = 0.2$
- Lead: $l = 8\,\mathrm{mm} = 0.008\,\mathrm{m}$
- Pitch: $p = 2\,\mathrm{mm}$
- Diameter: $d = 8\,\mathrm{mm}$
- Mean diameter: $d_m = d - \frac{p}{2} = 7\,\mathrm{mm}$

$$\begin{aligned} T &= \frac{Fd_m}{2} \cdot \left(\frac{l + \pi\mu d_m}{\pi d_m - \mu l} \right) \\ &= \frac{(300)(0.007)}{2} \cdot \left(\frac{0.008 + \pi(0.2)(0.007)}{\pi(0.007) - (0.2)(0.008)} \right) \\ &= 0.638\,\mathrm{N \cdot m} \end{aligned}$$

A.2 Z Axis Lead Screw Calculations

Assumptions:

- Load mass: $m = 1.86$ kg

Required lifting force:
$F = mg = 1.86 \cdot 9.81 = 18.25\,\mathrm{N}$
Required torque for lead screw to produce axial force:

$$\begin{aligned} T &= \frac{Fd_m}{2} \cdot \left(\frac{l + \pi\mu d_m}{\pi d_m - \mu l} \right) \\ &= \frac{18.25 \cdot 0.007}{2} \cdot \left(\frac{0.008 + \pi \cdot 0.2 \cdot 0.007}{\pi \cdot 0.007 - 0.2 \cdot 0.008} \right) \\ &= 0.0388\,\mathrm{N \cdot m} \end{aligned}$$

A.3 X and Y Axis Belt and Pulley Calculations

Assumptions:

- Mass: $m_x = 0.39\,\text{kg}$, $m_y = 1.02\,\text{kg}$
- Desired acceleration: $a = 5\,\text{m/s}^2$
- Coefficient of friction: $\mu = 0.2$

Required belt force along the X axis:

$$F_{\text{x,inertia}} = m_x \cdot a = 0.39 \cdot 5 = 1.95\,\text{N}$$

$$F_{\text{x,friction}} = \mu \cdot m_x \cdot g = 0.2 \cdot 0.39 \cdot 9.81 = 0.77\,\text{N}$$

$$F_{\text{x,total}} = F_{\text{x,inertia}} + F_{\text{x,friction}} = 1.95 + 0.77 = \boxed{2.72\,\text{N}}$$

Required pulley torque to produce belt force in the X direction:
A GT2 20T pulley has a diameter of:

$$d_p = \frac{t \cdot N}{\pi} = \frac{2 \cdot 20}{\pi} \approx 12.73\,\text{mm} = 0.01273\,\text{m}$$

Therefore, the required torque is:

$$T = F \cdot \frac{d_p}{2} = 2.72 \cdot \frac{0.01273}{2} = \boxed{0.0173\,\text{Nm}}$$

Required belt force along the Y axis:

$$F_{\text{y,inertia}} = m_y \cdot a = 1.02 \cdot 5 = 5.1\,\text{N}$$

$$F_{\text{y,friction}} = \mu \cdot m_y \cdot g = 0.2 \cdot 1.02 \cdot 9.81 = 2.00\,\text{N}$$

$$F_{\text{y,total}} = F_{\text{y,inertia}} + F_{\text{y,friction}} = 5.1 + 2.00 = \boxed{7.1\,\text{N}}$$

Required pulley torque to produce belt force in the Y direction:
A GT2 36T pulley has a diameter of:

$$d_p = \frac{t \cdot N}{\pi} = \frac{2 \cdot 36}{\pi} \approx 22.92\,\text{mm} = 0.02292\,\text{m}$$

Therefore, the required torque is:

$$T = F \cdot \frac{d_p}{2} = 7.1 \cdot \frac{0.02292}{2} = \boxed{0.08137\,\text{Nm}}$$

Appendix B

Mechanical Drawings

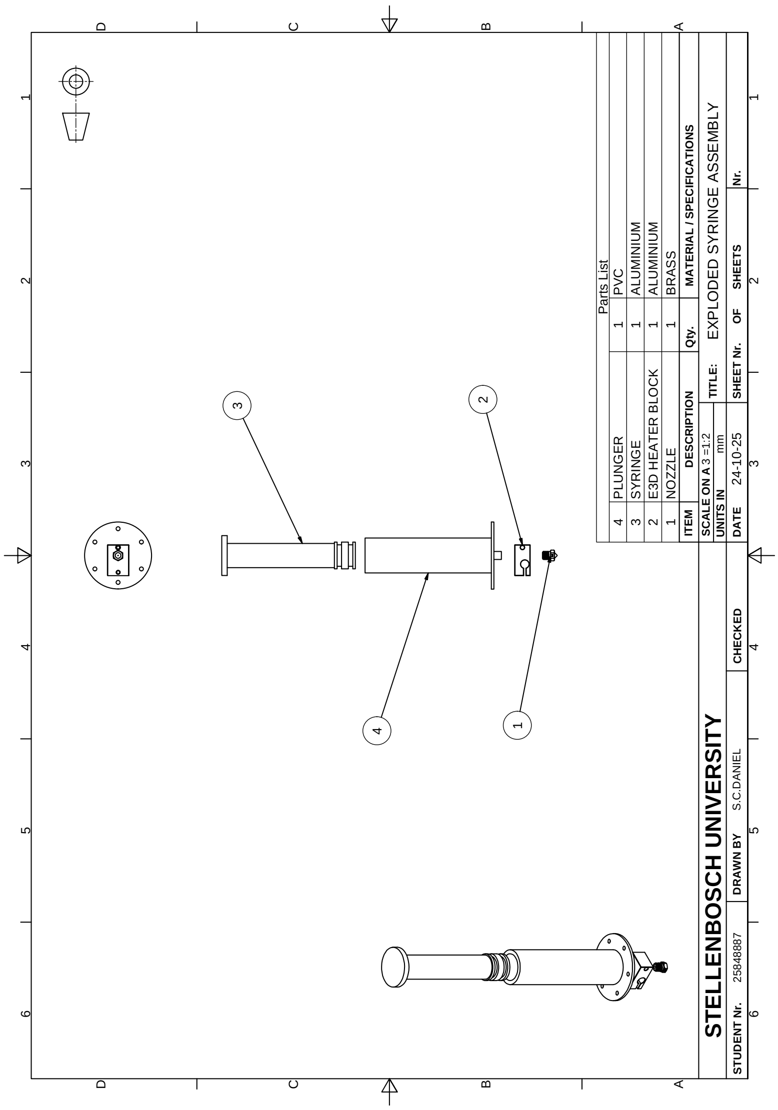


Figure B.1: Mechanical drawing of the exploded syringe assembly.

Appendix C

Embedded Hardware

This appendix shows the complete embedded system hardware design, including the 3D model, copper layer layouts, and schematic pages for the hydrogel 3D printer control circuit.

C.1 PCB Layout

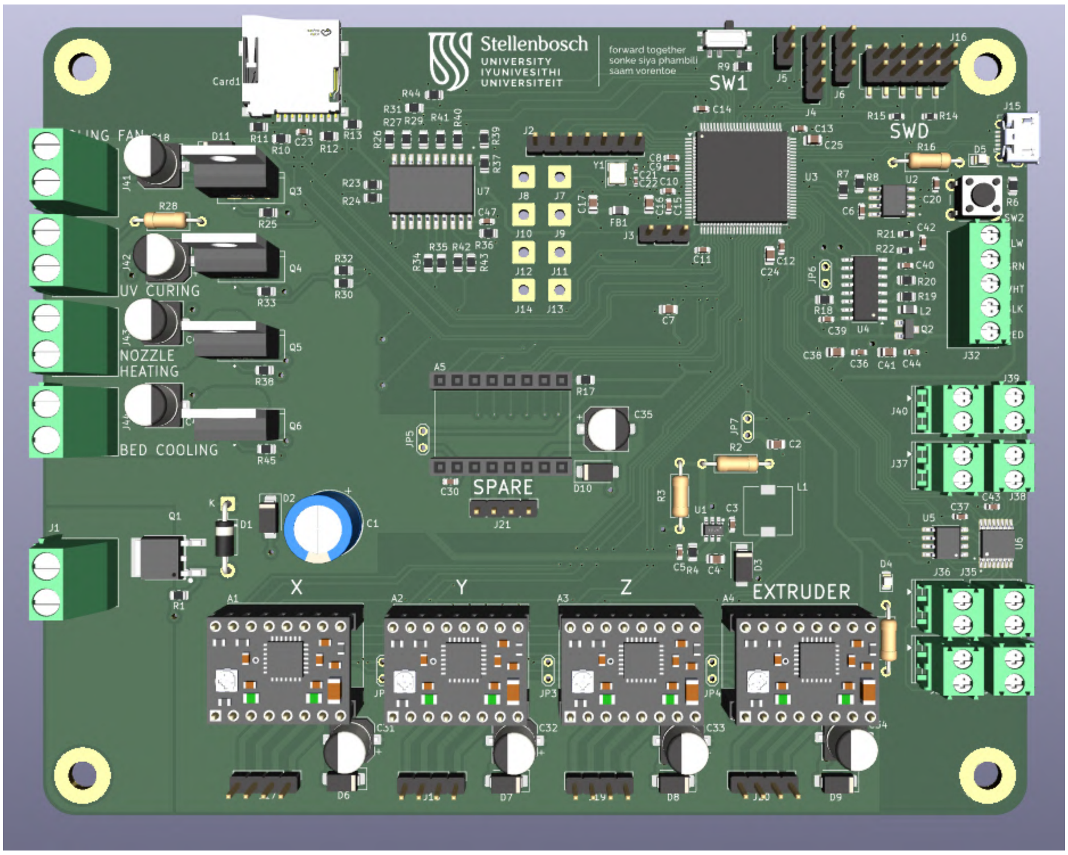


Figure C.1: 3D model of the assembled PCB illustrating overall component layout

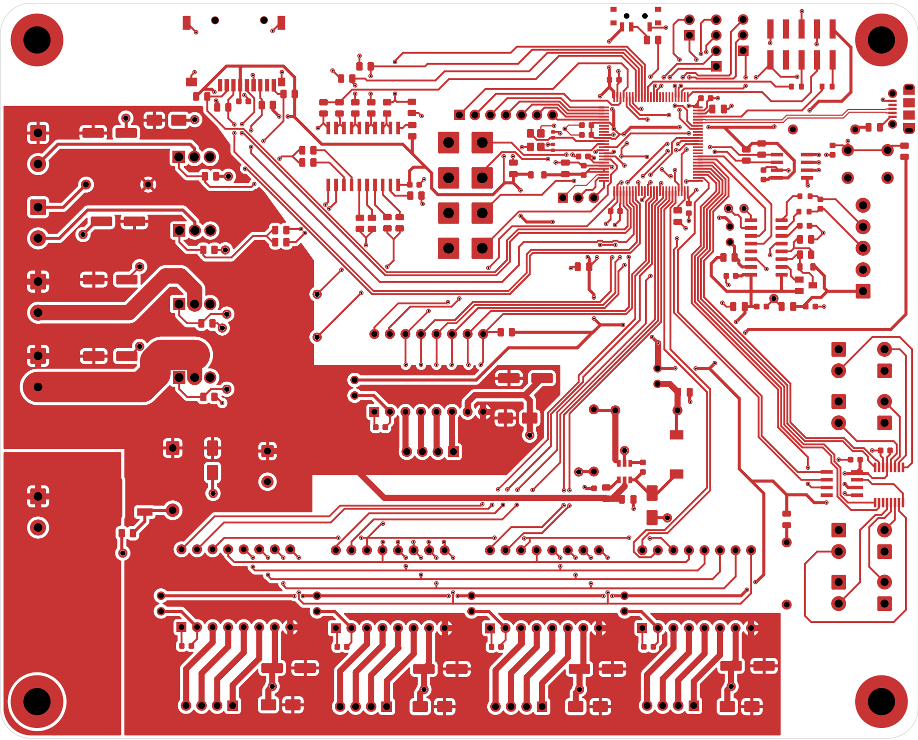


Figure C.2: Top layer layout showing signal routing and power traces.

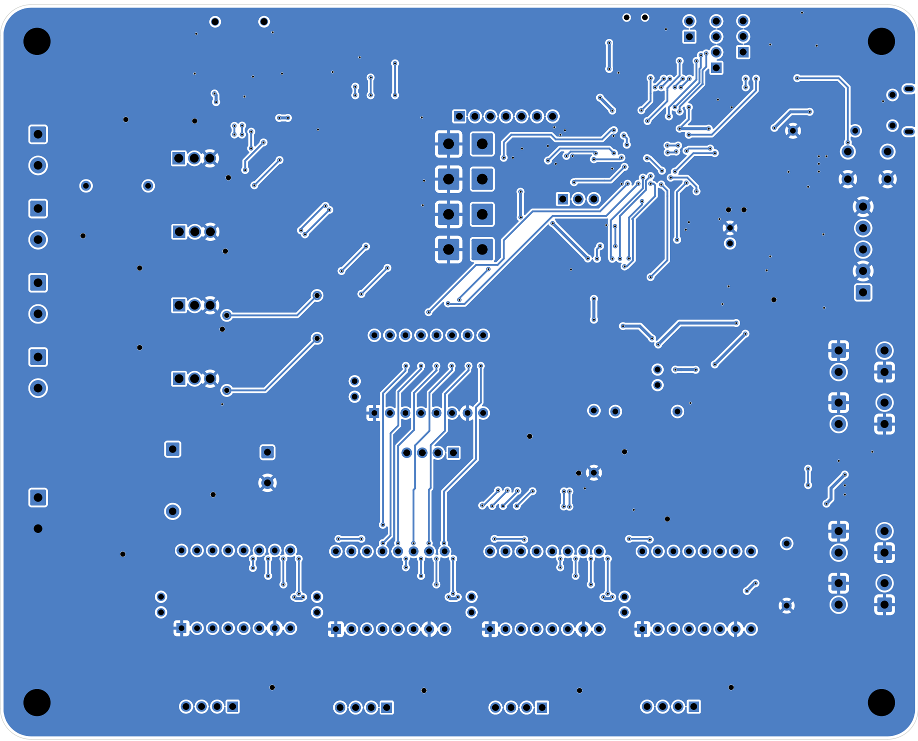


Figure C.3: Bottom layer layout showing ground planes, vias, and routing.

C.2 Circuit Schematics

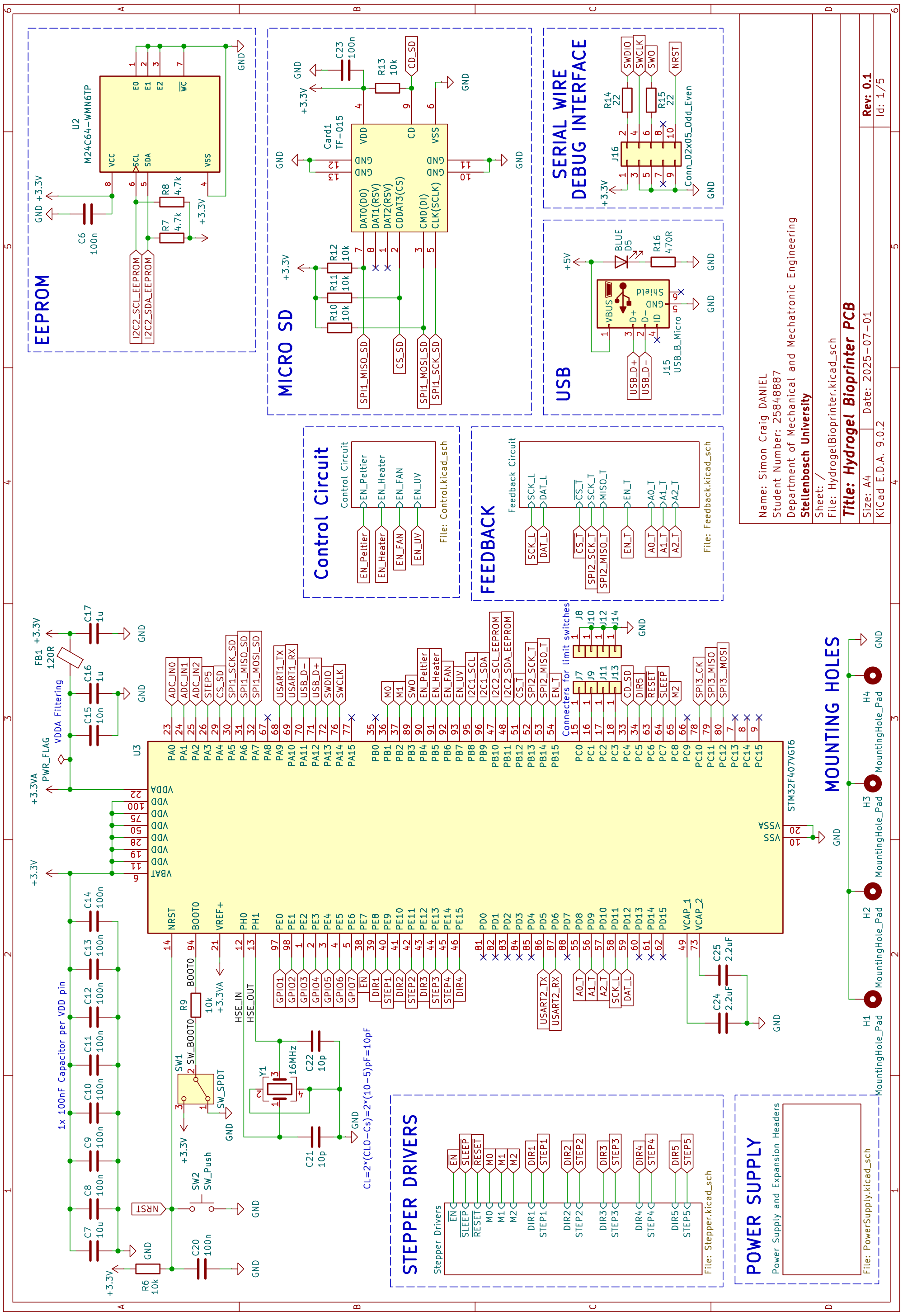


Figure C.4: Schematic overview of the bioprinter control board.

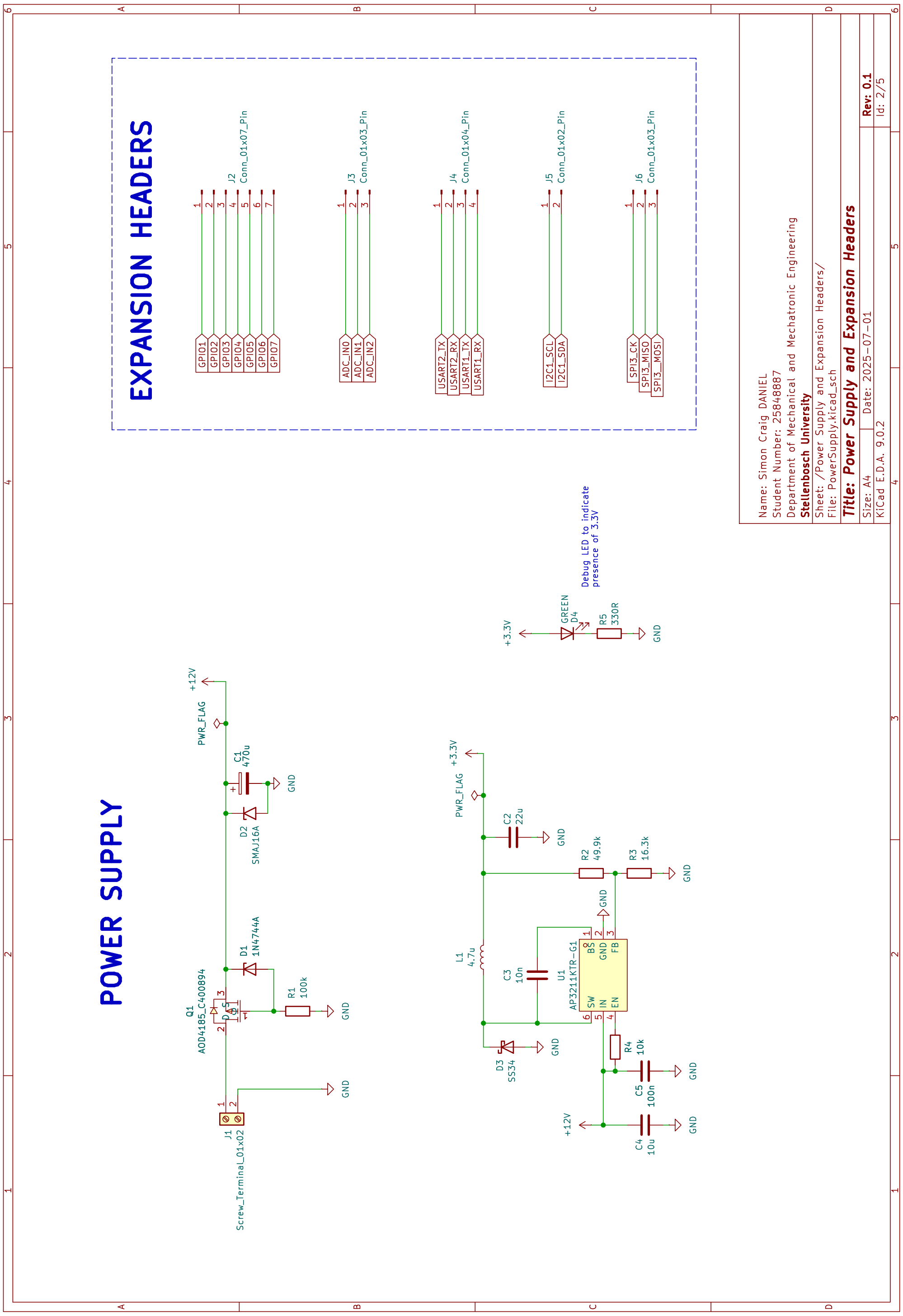


Figure C.5: Bioprinter power supply system schematic.

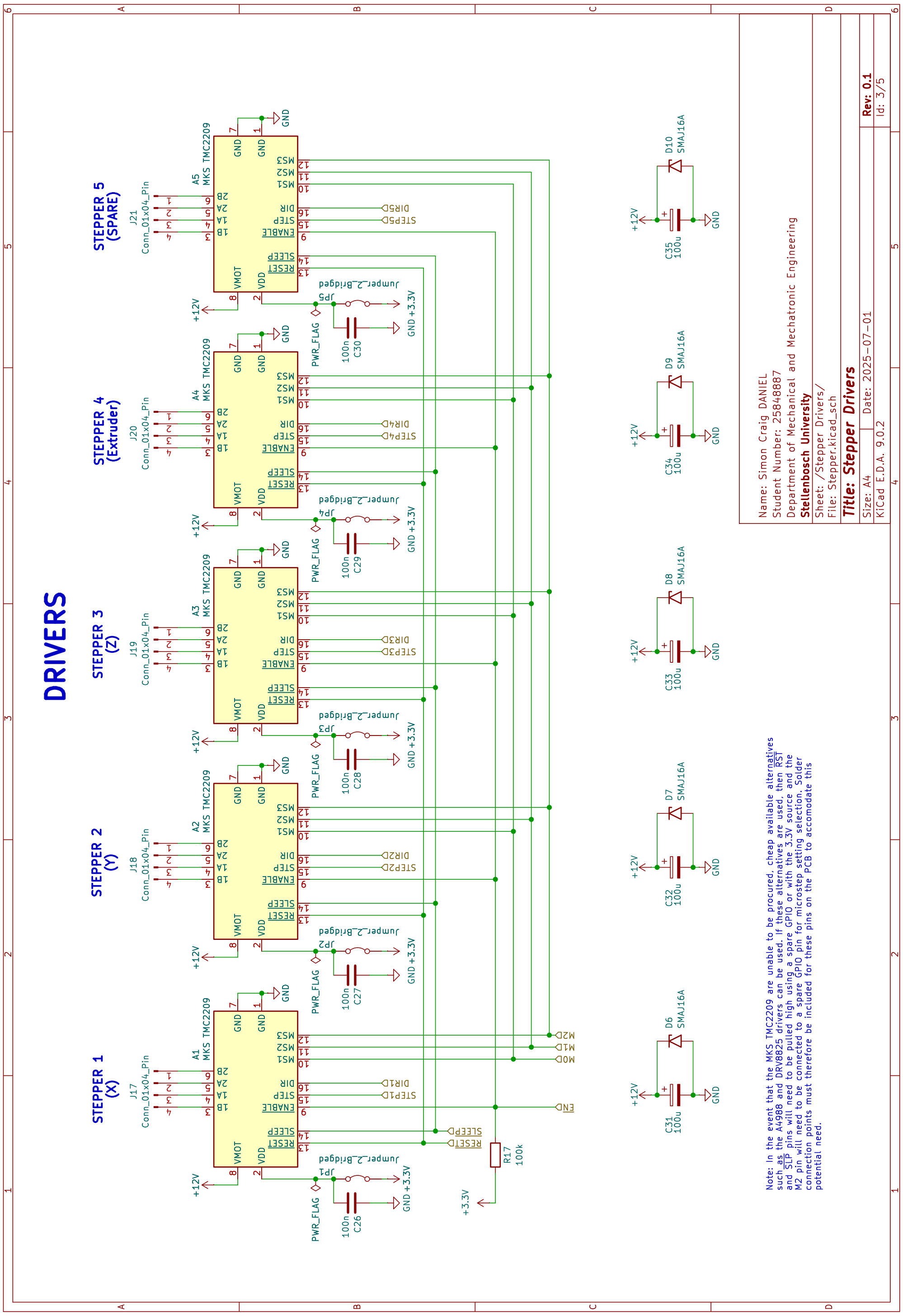


Figure C.6: Bioprinter stepper driver system schematic.

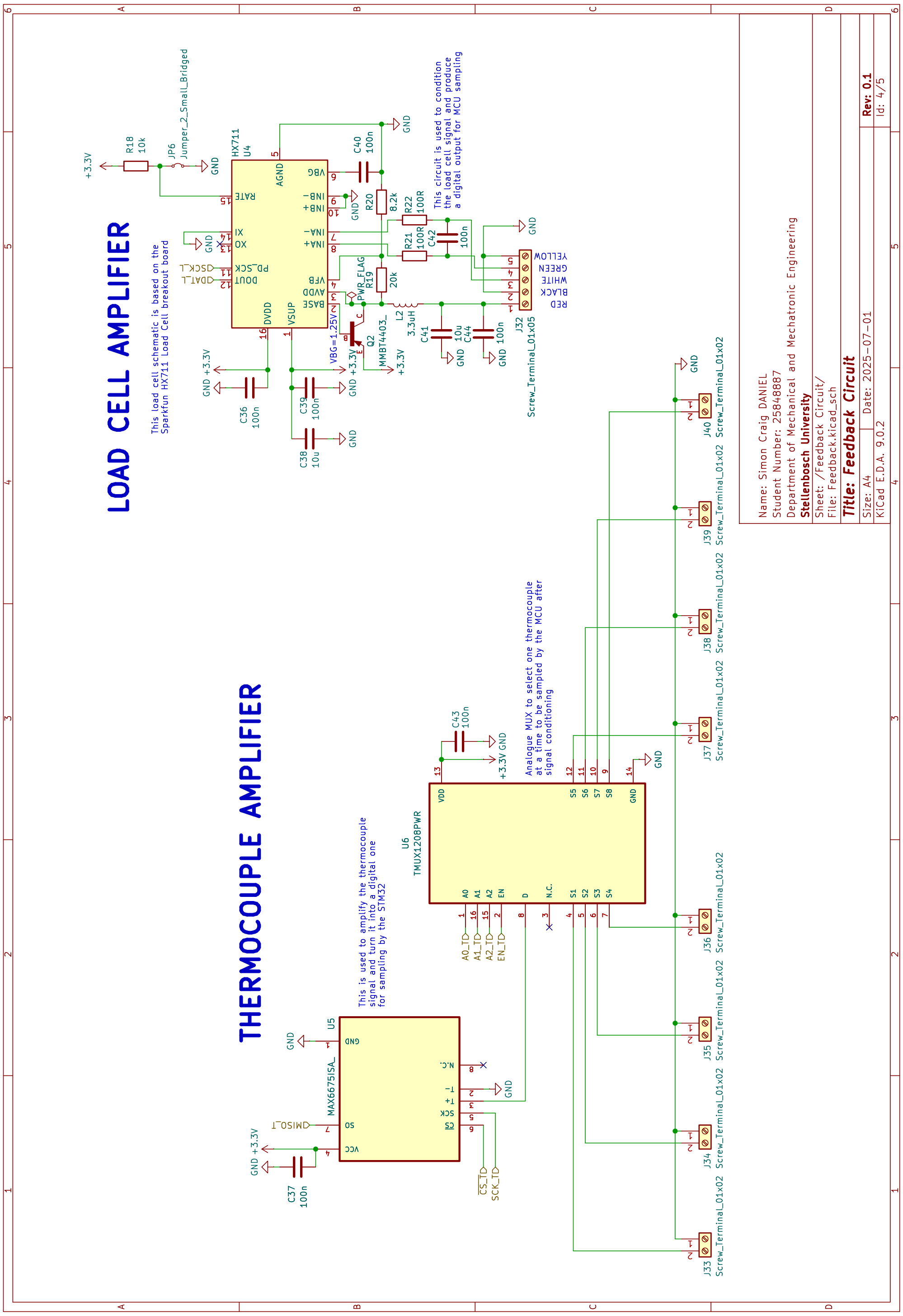


Figure C.7: Bioprinter sensor system schematic.

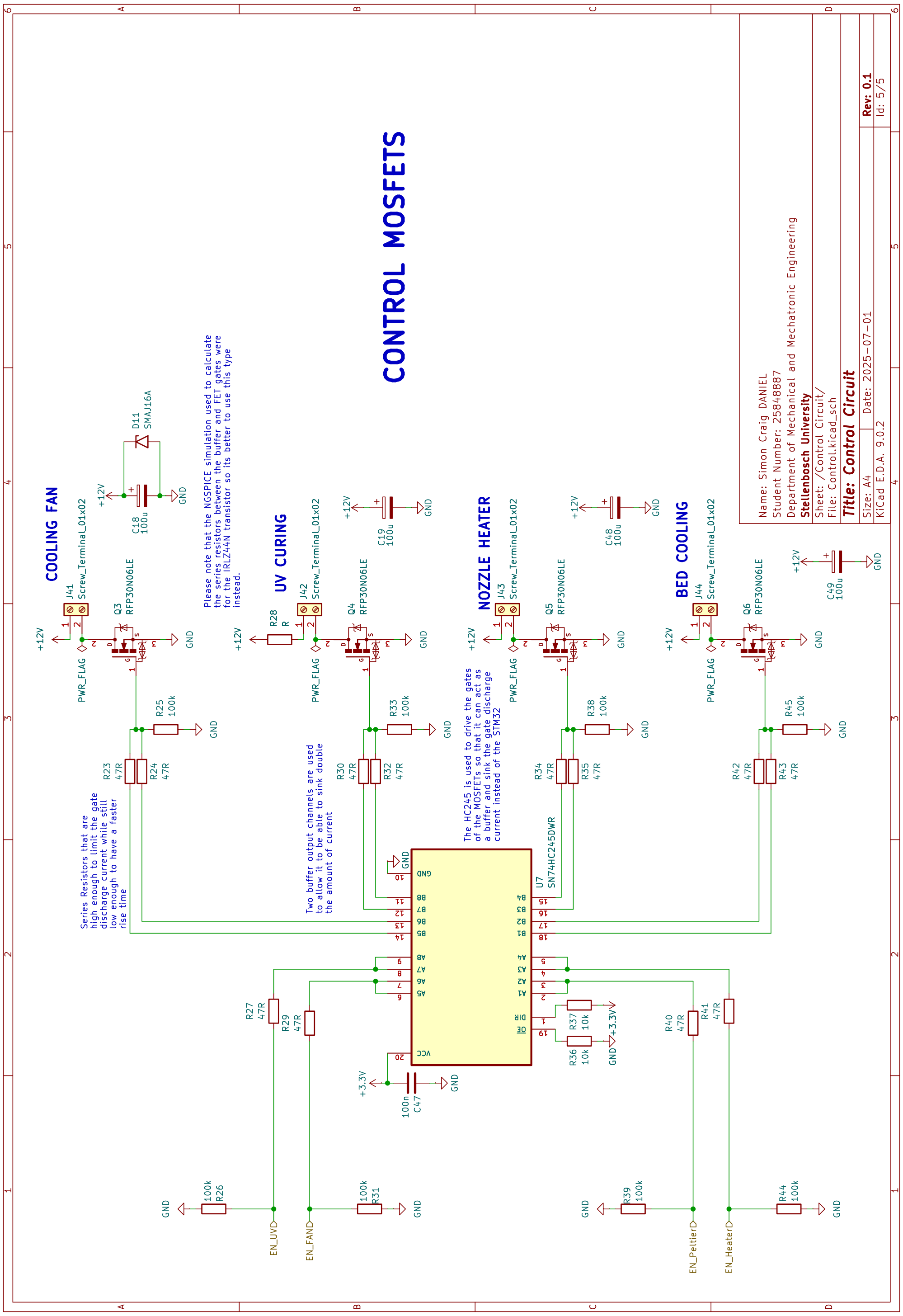


Figure C.8: Bioprinter control circuit schematic.

Appendix D

Experimental Calibration

D.1 Calibration Test Matrices

Table D.1: Proposed Test Matrix for Optimal Nozzle height

Test No.	Nozzle Height (mm)
1	0.2
2	0.4
3	0.6
4	0.8
5	1.0

Table D.2: Proposed Test Matrix for Straight Line Experiment

Test No.	Printing Speed (mm/s)	Extrusion Speed (mm/s)
1	20	0.2
2	20	0.3
3	20	0.4
4	20	0.5
5	25	0.2
6	25	0.3
7	25	0.4

8	25	0.5
9	30	0.2
10	30	0.3
11	30	0.4
12	30	0.5
13	35	0.2
14	35	0.3
15	35	0.4
16	35	0.5

Table D.3: Proposed Test Matrix for Curved Line Experiment

Test No.	**Line Radius (mm)**
1	5
2	15
3	25
4	35
5	45

Table D.4: Proposed Test Matrix for Optimal Spacing of Adjacent Lines

Test No.	**Line Spacing (mm)**
1	3.0
2	3.5
3	4.0
4	4.5
5	5.0
6	5.5

Table D.5: Proposed Test Matrix for Optimal Layer Stacking Offset

Test No.	**Layer Offset (mm)**
1	1.0
2	1.5
3	2.0

D.2 Measured Results

Table D.6: Calibration Phase 1 - Nozzle Height Test Results

Test No.	**Nozzle Height (mm)**	**Length (mm)**	**Mean Width (mm)**	**Std Dev Width (mm)**
1	0.2	88	2.9	0.71
2	0.4	94	3.0	0.33
3	0.6	93	3.0	0.53
4	0.8	92	3.0	0.52
5	1.0	90	2.8	0.47

Table D.7: Calibration Phase 2 - Straight Line Test Results

Test No.	**Print Speed (mm/s)**	**Extrusion Speed (mm/s)**	**Length (mm)**	**Mean Width (mm)**	**Std Dev Width (mm)**
1	20	0.2	93	3.0	0.27
2	20	0.3	94	3.6	0.57
3	20	0.4	99	4.2	0.32
4	20	0.5	101	4.5	0.34
5	25	0.2	85	3.2	0.46
6	25	0.3	90	3.5	0.63

Table D.7: Calibration Phase 2 - Straight Line Test Results

Test No.	Print Speed (mm/s)	Extrusion Speed (mm/s)	Length (mm)	Mean Width (mm)	Std Dev Width (mm)
7	25	0.4	94	3.8	0.26
8	25	0.5	98	4.1	0.56
9	30	0.2	87	2.3	0.40
10	30	0.3	97	2.9	0.46
11	30	0.4	98	3.7	0.45
12	30	0.5	99	4.5	0.53
13	35	0.2	92	2.2	0.53
14	35	0.3	95	2.9	0.74
15	35	0.4	97	3.9	0.57
16	35	0.5	98	4.1	0.79

Table D.8: Calibration Phase 3 - Curved Arc Test Results

Test No.	Radius (mm)	Arc Angle (°)	Mean Width (mm)	Std Dev Width (mm)
1	5	352	4.5	1.16
2	15	360	4.6	0.28
3	25	357	4.7	0.27
4	35	354	4.5	0.38
5	45	357	4.5	0.50

Table D.9: Calibration Phase 6 - Repeatability Test Results

Test No.	Length (mm)	Mean Width (mm)	Std Dev Width (mm)
1	98	4.5	0.21

Table D.9: Calibration Phase 6 - Repeatability Test Results

Test No.	Length (mm)	Mean Width (mm)	Std Dev Width (mm)
2	98	4.5	0.35
3	99	4.3	0.26
4	98	4.7	0.30
5	99	4.6	0.45

D.3 Visual Record of Printed Samples

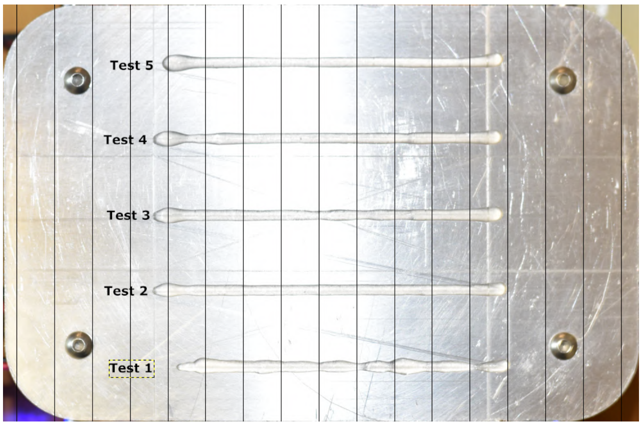


Figure D.1: Trial 1 — nozzle height test

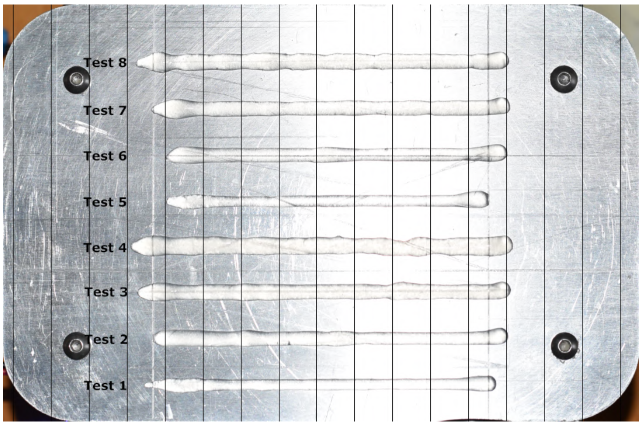


Figure D.2: Trial 2a — straight line test

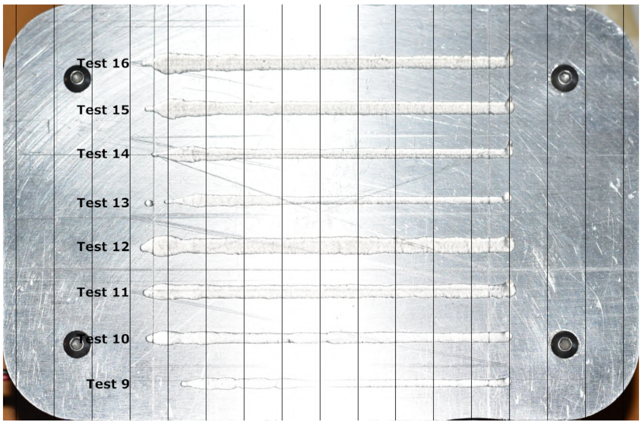


Figure D.3: Trial 2b — straight line test

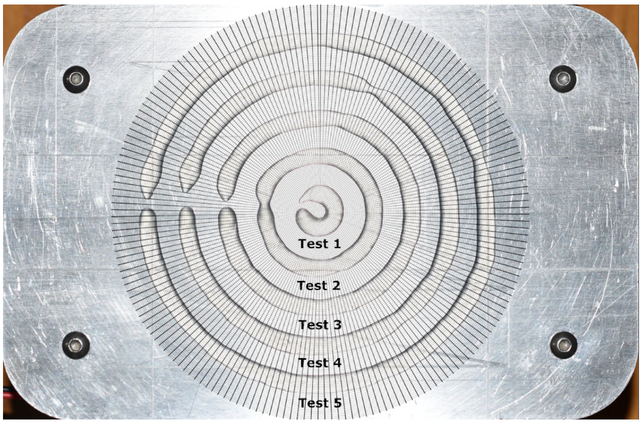


Figure D.4: Trial 3 — circle test

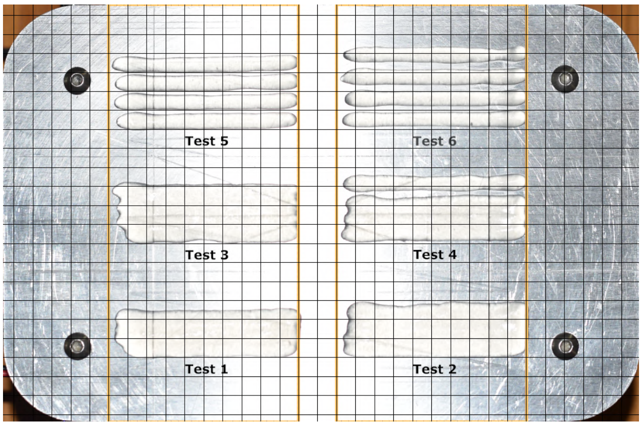


Figure D.5: Trial 4 — adjacent lines test

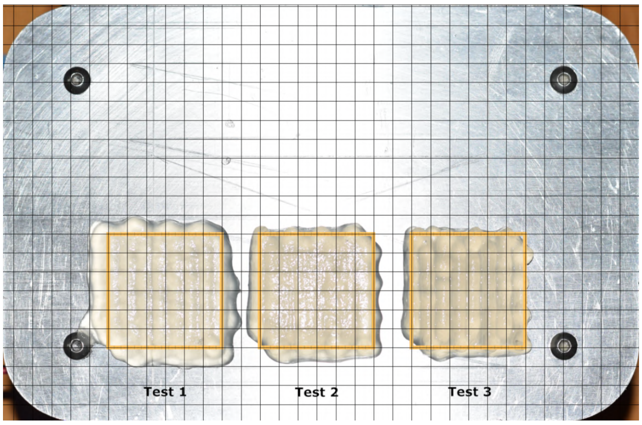


Figure D.6: Trial 5 — stacked layer test

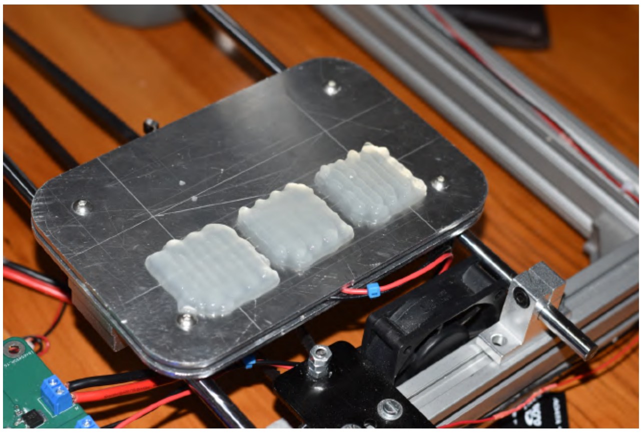


Figure D.7: Trial 5 — angled view

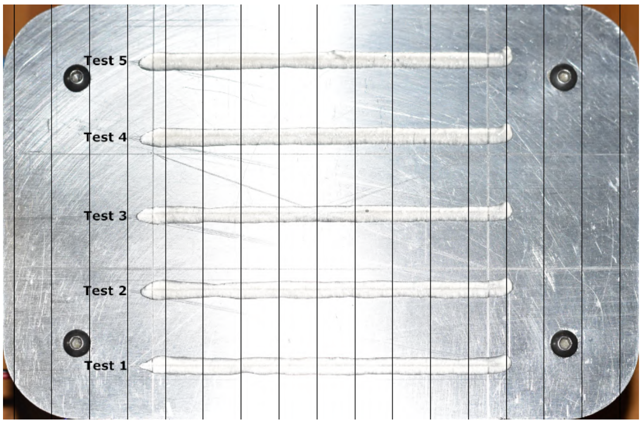


Figure D.8: Trial 6 — repeatability test

Appendix E

Techno-economic Analysis

This section discusses the technical and economic aspects and impacts of the project. It covers time management, the project budget, technical impact, return on investment, and the project's potential for commercialisation.

E.1 Planning and Time Management

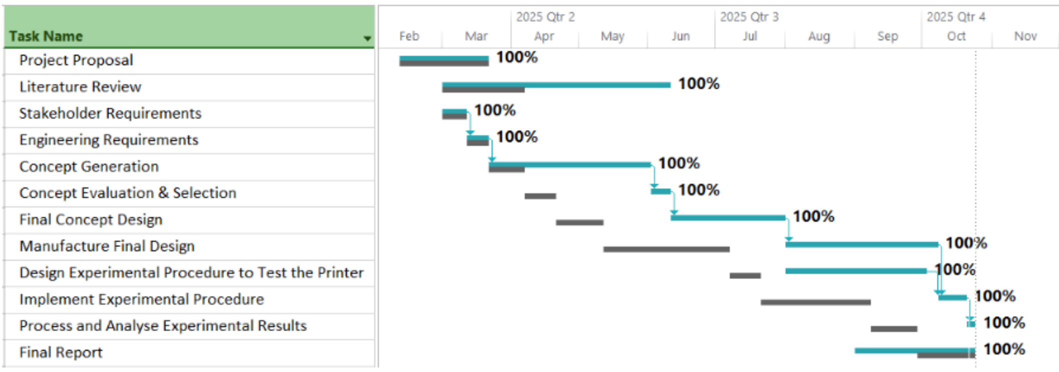


Figure E.1: Project Gantt Chart

The Gantt chart in Figure E.1 compares the predicted baseline duration (grey) of each project activity with the actual timeline (blue) in which the tasks were completed. It can be seen that the literature review extended over a significantly longer period than initially planned, due to the high workload from other modules during the first semester. A second major deviation occurred during the concept generation phase, which exceeded its planned duration because of an overly optimistic initial time estimate. The remaining project activities – including completing the design process, manufacturing, assembling, and conducting the experimental work – had to be completed within a shorter time frame to meet the final project deadline of 24 October 2025.

E.2 Budget

Table E.1: Estimated Project Budget

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW Labour		MMW Material	Total
	hr	R	R	R	R	hr	R	R	R
Literature Review	48	21 600							21 600
Stakeholder Requirements	12	5 400							5 400
Engineering Requirements	12	5 400							5 400
Concept Generation	24	10 800							10 800
Concept Evaluation and selection	18	8 100							8 100
Final Concept Design	36	16 200							16 200
Manufacture Final Design	120	54 000	1 000	6 000	1 500	24	7 200	2 000	71 700
Design Experimental Procedure to Test the Printer	18	8 100							8 100
Implement Experimental Procedure	48	21 600	1 000		100				22 700
Process and Analyse Experimental Results	24	10 800							10 800
Final Report	120	54 000							54 000
Total	480	216 000	2 000	6 000	1 600		7 200	2 000	**234 800**

Table E.2: Actual Project Cost

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW Labour		MMW Material	Total
	hr	R	R	R	R	hr	R	R	R
Literature Review	45	20 250							20 250
Stakeholder Requirements	12	5 400							5 400
Engineering Requirements	12	5 400							5 400
Concept Generation	36	16 200							16 200
Concept Evaluation and selection	10	4 500							4 500
Final Concept Design	72	32 400							32 400
Manufacture Final Design	100	45 000	450	5 000	7 910	10	5 500	240	64 100
Design Experimental Procedure to Test the Printer	18	8 100							8 100
Implement Experimental Procedure	30	13 500	4 000						17 500
Process and Analyse Experimental Results	20	9 000							9 000
Final Report	100	45 000							45 000
Total	455	204 750	4 450	5 000	7 910		5 500	240	**227 850**

The predicted budget and estimated final costs are shown in Tables E.1 and E.2. The actual cost of R227 850 was slightly below the initially predicted budget of R234 800. The largest discrepancies between the planned and actual expenses occurred in the running, capital, and workshop cost categories. These differences occurred because the final design was completed after the proposed budget had been estimated. Therefore, many required components that needed to be purchased or manufactured were not yet identified. In addition, the running cost estimate did not accurately account for electricity consumption and unplanned consumables such as lubricant.

The cost of component procurement, whether manufactured by the university or externally sourced, is summarised in Table E.3. This table reflects the total cost of developing the complete prototype, excluding the engineering time required for the system design.

Table E.3: Prototype Costs

Category	Cost (R)
University Orders	6 604.19
Personal Purchases	1 234.00
Workshop Labour	5 740.00
3D Printing	71.87
Total	**13 649.96**

E.3 Technical Impact

The development of the hydrogel 3D printer provides a meaningful technical contribution to biomedical and tissue engineering research. The project demonstrates that repeatable hydrogel extrusion can be achieved using a cost-effective bioprinting system. On a technical level, the functional prototype serves as a flexible research tool for studying flow behaviour of unknown hydrogels. Building on the first iteration of this project – done in 2024 – this prototype integrates embedded control, modular hardware, and tunable parameters. It has also lowered the cost barrier for local tissue engineering research and established a stronger foundation for continued bioprinter development within the university.

E.4 Return on Investment

The time and money that was invested into this project resulted in a functional research platform. In the short term, the bioprinter serves as a means of characterising different unknown gels. In the long term, the system offers a low-cost alternative to imported high-cost bioprinters. Further development of this project by future students can refine the prototype into a laboratory-grade bioprinter with temperature control and closed-loop pressure feedback. The project therefore provides a high return on investment through its affordability, adaptability, and potential use as a research tool.

E.5 Potential for Commercialization

The project outcome is a functional prototype of a hydrogel 3D printer that shows potential for commercialization as an affordable and modular bioprinter for research institutions. Current commercial bioprinters, like the BIO X series, are not accessible for many laboratories due to their high costs, creating a market opportunity for more affordable alternatives. The open-source firmware and modular design makes the system adaptable to a wide range of hydrogels. With further refinement in sterility and addition of closed-loop syringe temperature and pressure control, an enclosure design, and user interface, the prototype can be developed to serve as a low-cost bioprinter for creating cell-laden scaffolds and characterising hydrogels for universities and other research institutions.

Appendix F

Risk Analysis and Safety Procedures

Safety Overview

The mechanical and embedded system design, assembly and testing of the hydrogel 3D printer were performed at home, in a personal workspace, rather than in an official university laboratory or workshop. The project was conducted using a low-voltage electronic system and non-hazardous materials, and therefore presented minimal risks to both the user and equipment. Safety considerations, associated with the project, focused mainly on the mechanical, electrical, and thermal hazards associated with component assembly, soldering, and system handling.

The workspace was kept neat, tidy and ventilated to reduce the probability of tripping, remove solder fumes, and maintain easy access to equipment. Hazards such as the sharp edges of cut aluminium extrusions were eliminated by filing all edges after cutting. In cases where risks could not be completely eliminated, such as the presence of hot surfaces and nipping points, careful handling procedures were followed. This included keeping hands clear of the printer whenever in operation during the calibration stage and avoiding contact with heated components such as the nozzle and heat sinks until sufficient cooling time had elapsed after the printer was powered-down. To ensure the safe operation of the electronic system, protective measures were also implemented, such as limiting stepper driver current and verifying circuit connections with a multimeter before supplying power to the system.

Activity based Risk Assessment

A risk assessment for the project is provided in Table F.1, which identifies the potential risks associated with each of project activity. The table also lists the corresponding steps to mitigate the likelihood and severity of each hazard to ensure personal safety (P) and equipment safety (E).

Table F.1: Risk assessment for the hydrogel 3D printer's development

Activity	**Risk**	**Risk Type**	**Mitigating Steps**
Cutting aluminium	Cuts from sharp edges or saw blade	P	File all cut edges; clamp parts securely during cutting.
Assembling the printer	Thread stripping or cracking of 3D printed parts	E	Avoid over-tightening bolts.
Soldering	Burns and inhalation of fumes	P	Solder in a ventilated area; keep the iron in its stand when not in use.
PCB assembly	Short circuits	E	Verify polarity of components; test for continuity; avoid over-soldering joints.
Operating motors	Excessive current draw of drivers	E	Set current limits on stepper drivers.
Operating the printer	Pinch points between moving parts	P	Keep hands clear of moving parts during operation.
Hot surfaces	Burns from heated nozzle and heat sinks	P	Keep hands clear of heated parts during operation; allow adequate cooling time after power-down before handling.
Circuit handling	Static discharge	E	Disconnect power before handling PCB; ensure correct breakout driver orientation.

Appendix G

Responsible Use of Resources and End-of-Life Strategy

Responsible Use of Resources

The hydrogel bioprinter was developed with the goal of using efficient materials and reusable components. The majority of the mechanical system consists of standard parts, including aluminium extrusions, TR8x8 lead screws, GT2 belts and pulleys, NEMA17 stepper motors, and linear bearings, which can all be reused in future student projects. 6 small PLA parts were 3D printed (4 feet and 2 limit switch mounts), resulting in minimal filament waste. The printer also includes several custom aluminium components (such as the syringe and bed plate) to ensure durability, most of which are made from aluminium plates to minimize machine waste.

The custom PCB was designed to maximise modularity, re-usability, and upgradability, by incorporating 5 stepper ports with breakout driver headers (StepStick form factor), 12 V switching ports, EEPROM storage, USB and SD card interface, and various breakout pins ensuring that the circuit board can be adapted across many different embedded systems applications. The experimental phase used store-bought agar powder as a hydrogel surrogate gel that is safe, biodegradable, inexpensive, and did not require a special disposal method.

End-of-Life Strategy

At the end of the bioprinter's operational life, it can be disassembled into its mechanical and electronic components. The aluminium extrusions, motors, and other standard parts can be reused in future university projects or recycled through standard metal recycling facilities (along with the rest of the custom aluminium components). The PCB can be re-purposed to operate in other mechatronic systems or responsibly disposed of as electronic-waste and the PLA components can be recycled. The bioprinter's modular design ensures that its components can be repaired, upgraded, or replaced, thereby extending the system's operational life and minimising environmental impact.

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